SDITO LAB

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Assignment 08

Problem Statement : Deploy a project from a local machine to GitHub and vice versa.

Aim

We need to deploy our local machine project into GitHub repository by command line interface by **git bash.** Also copy a GitHub repository project to my local machine.

Requirements

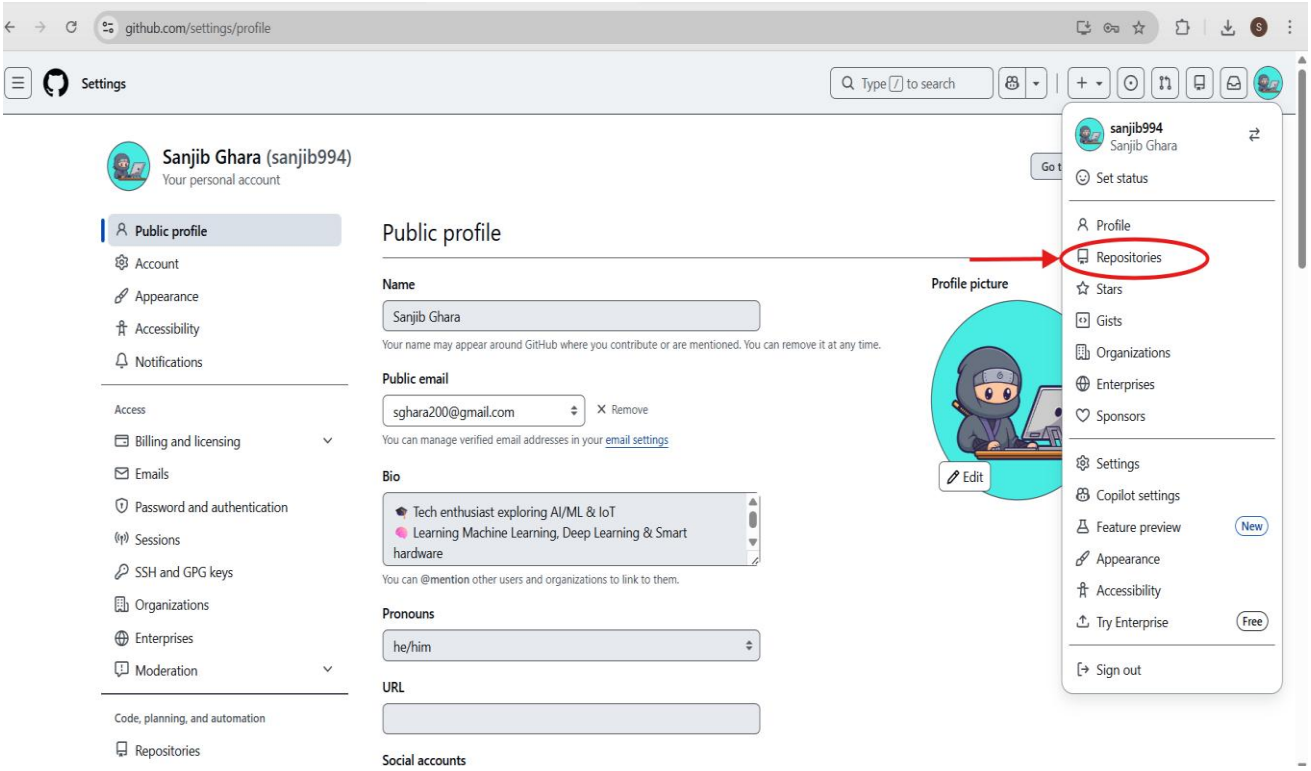
- GitHub account
- GitHub repo
- Git bash app
- Some commands
  I. git config --global --unset-all user.name
  II. git config --global user.name "sanjib994"
  III. git config --global user. Email sghara200@gmail.com
  IV. git config --global –list
  V. git init
  VI. git add .
  VII. git status
  VIII. git commit -m “”
  IX. git remote add origin <path>
  X. git push -u origin master

Solution:

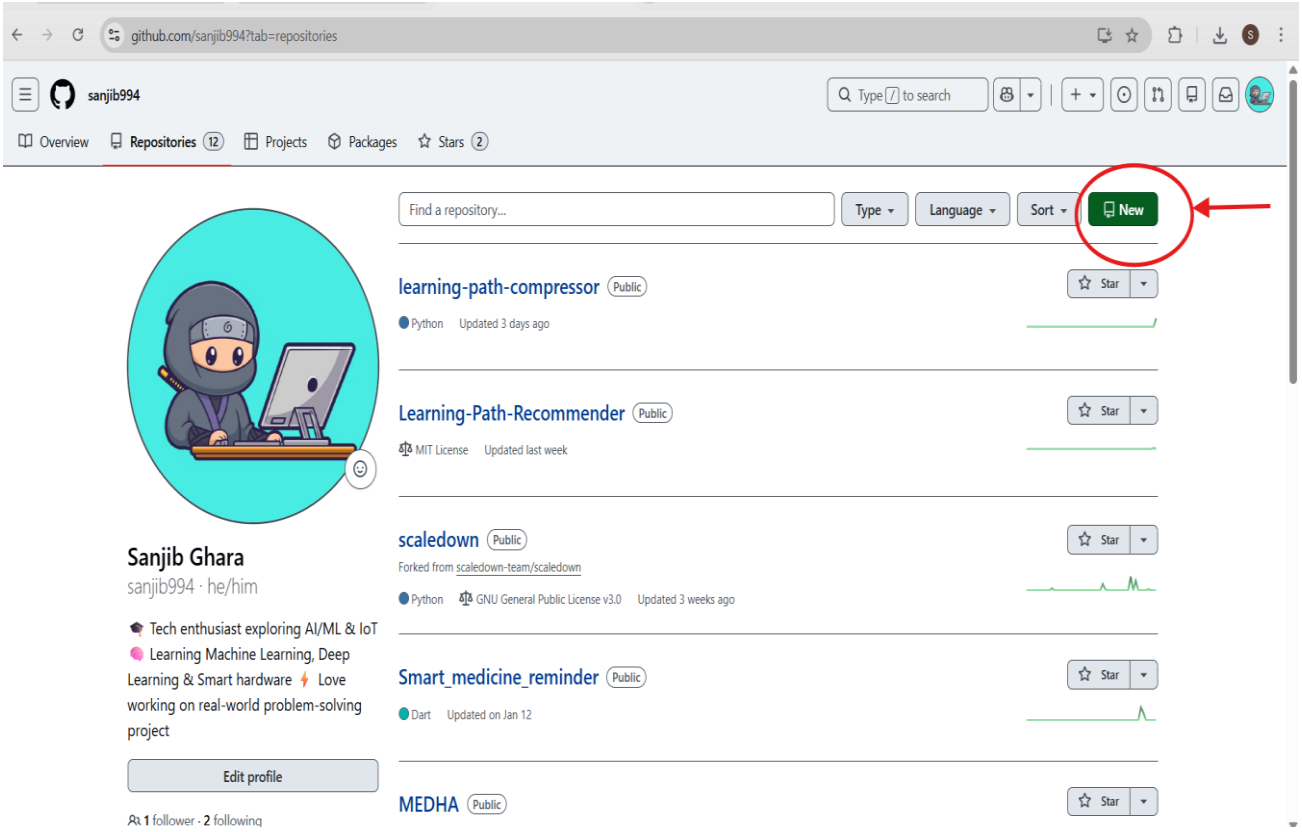
Step 1: at first, we need to open our GitHub account.

Step 2: then go to dashboard section.

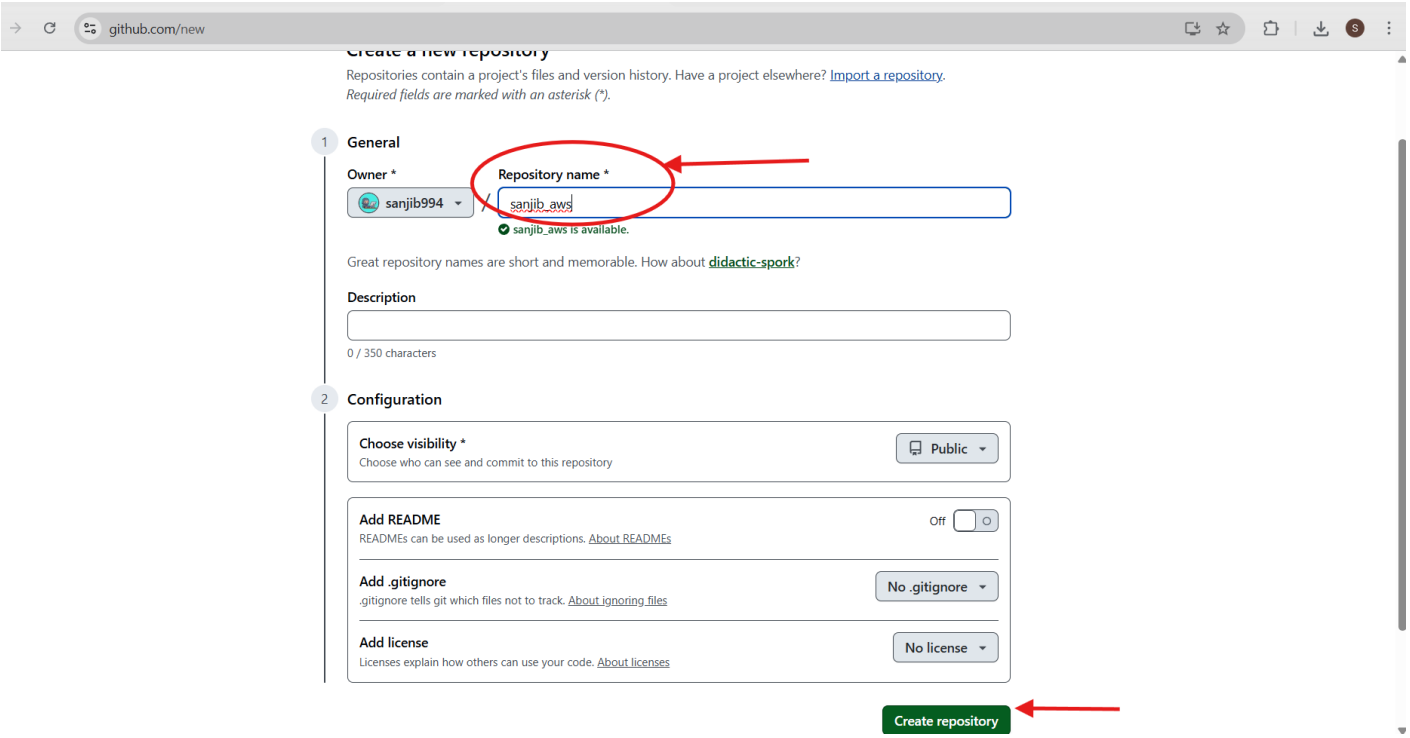
Step 3: after that click on repositories.



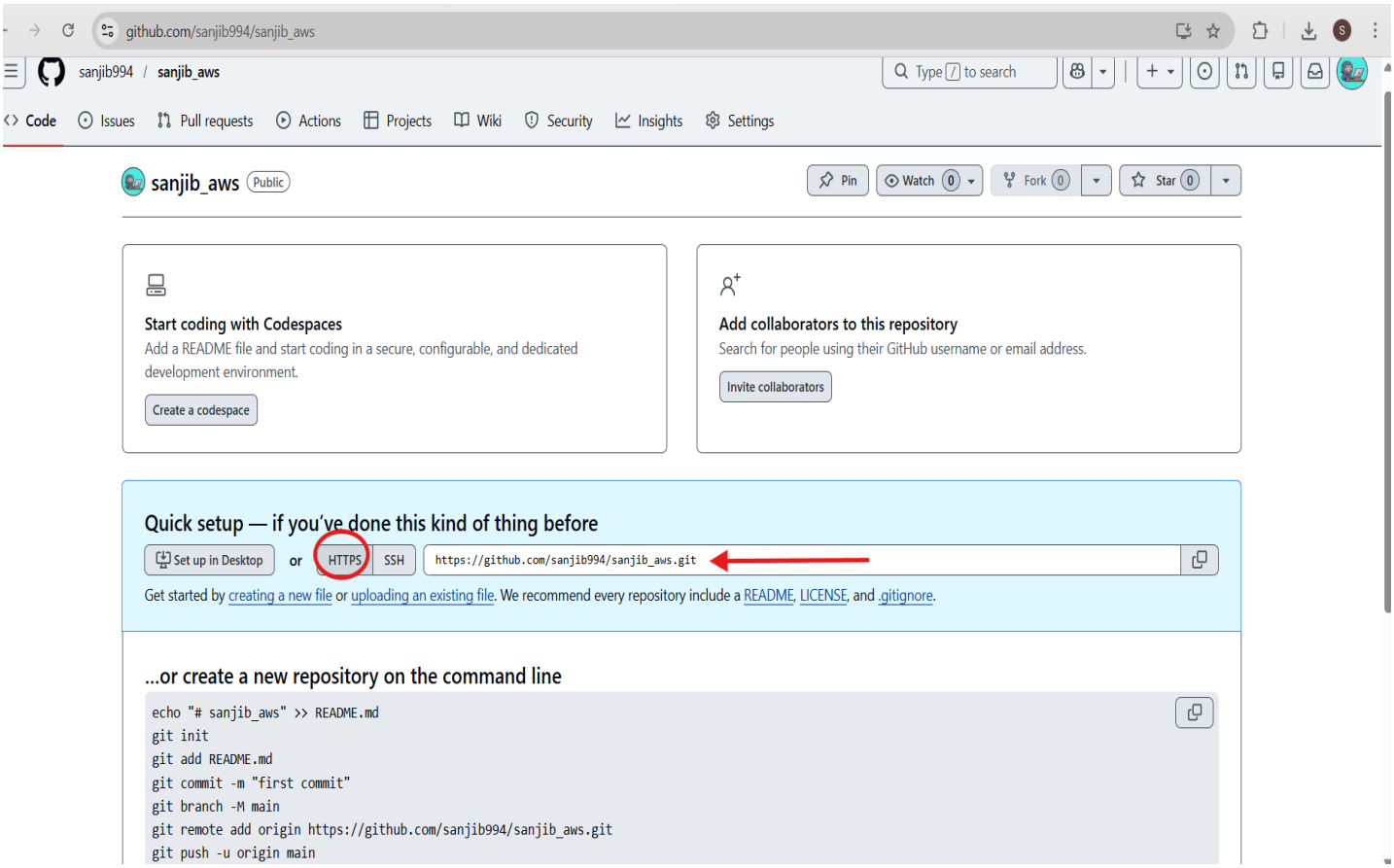
Step 5: now for create new repository we need to click on "NEW".



Step 6: now we need to give a repository name. then click on "create repository".

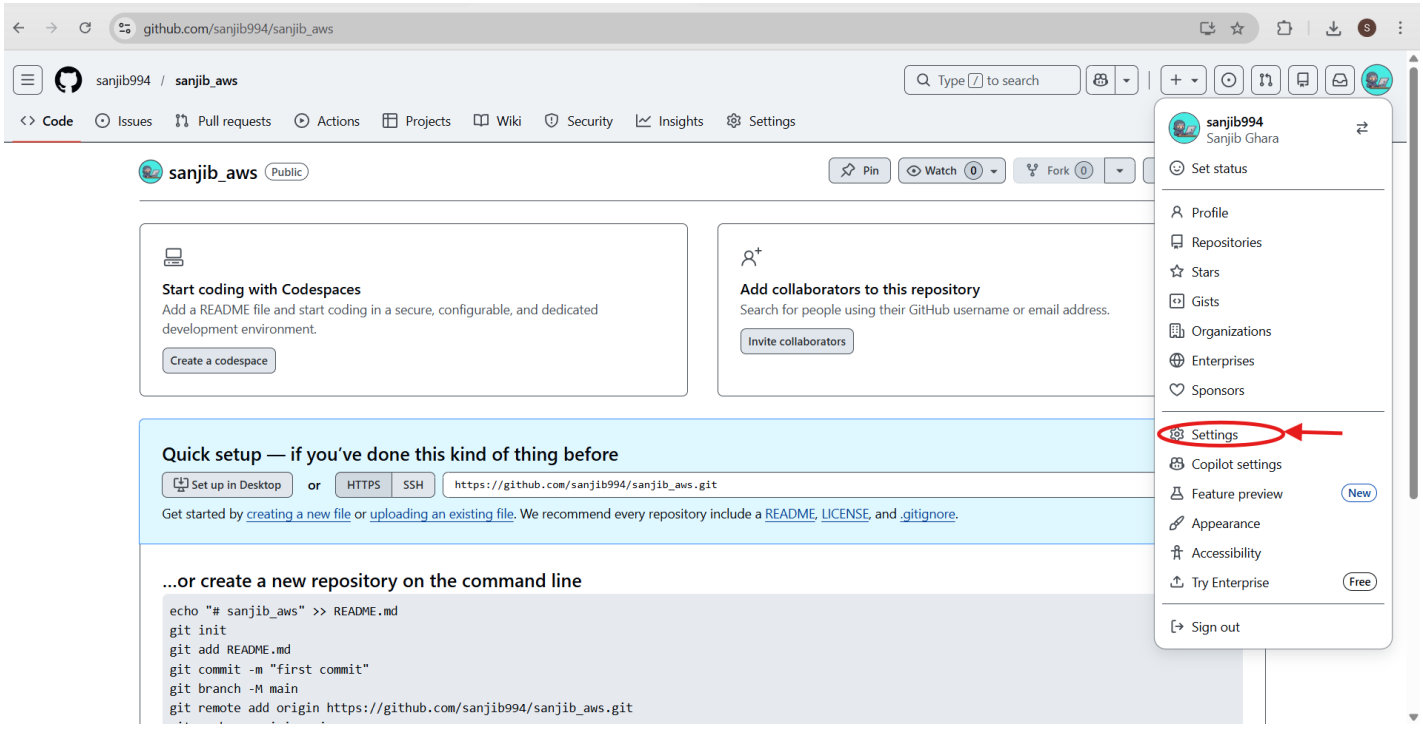


Step 7: now we need to copy the "HTTPS" link from my repo for future use.

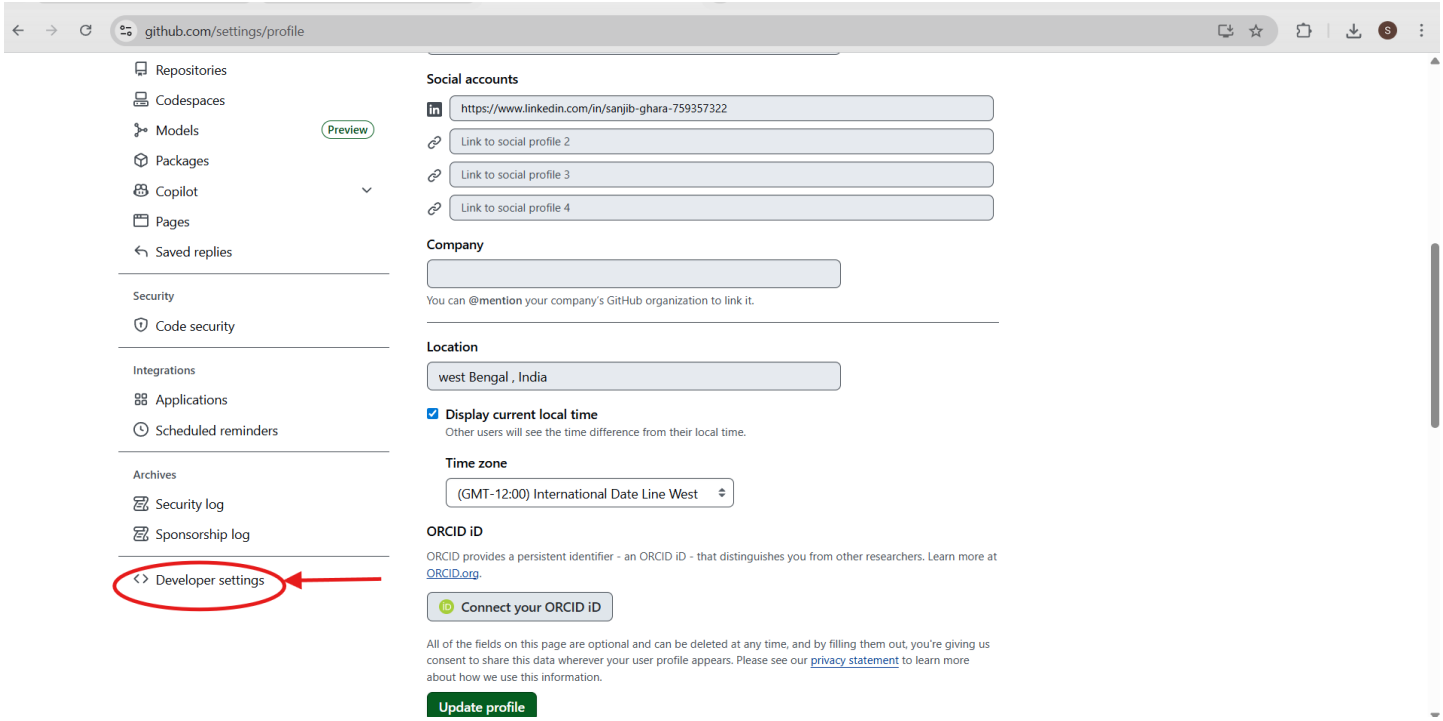


Step 8: now we need to create a classic token, which is useful for our any work on our repository.

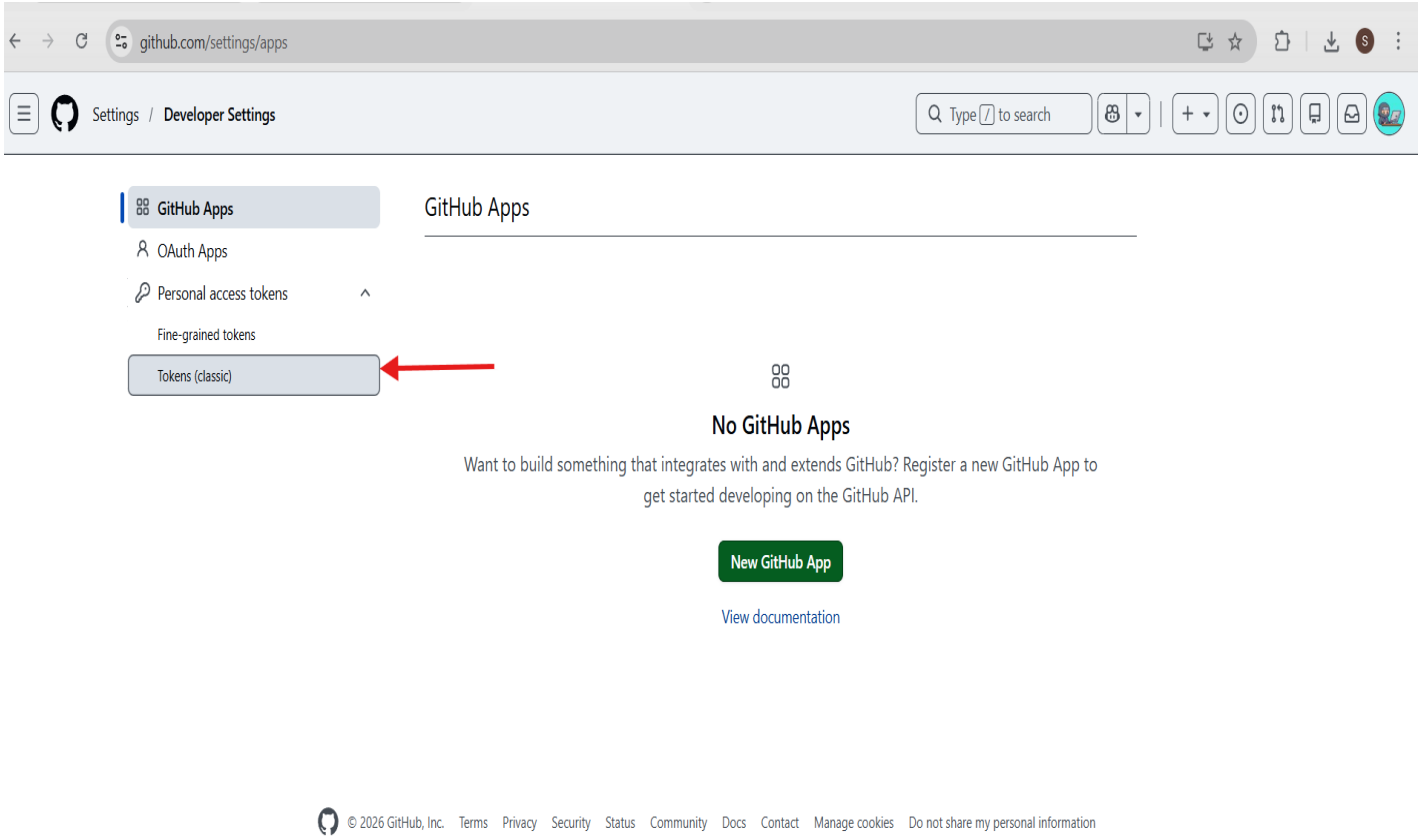
Step 9: to create a token, at first, we need to go to the settings option of GitHub profile.



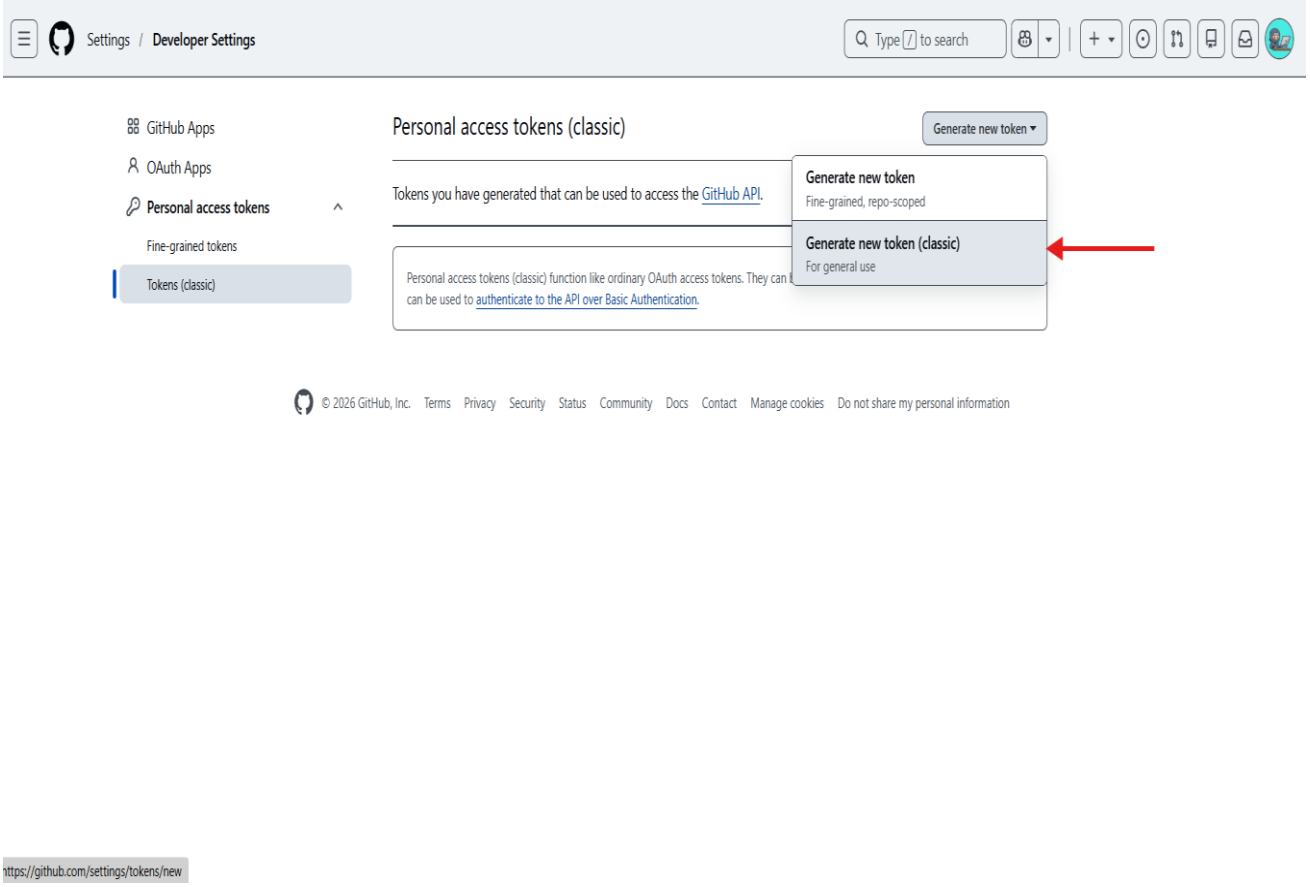
Step 10: after that we go to the bottom side of settings, where we find an option call "DEVELOPER SETTING".



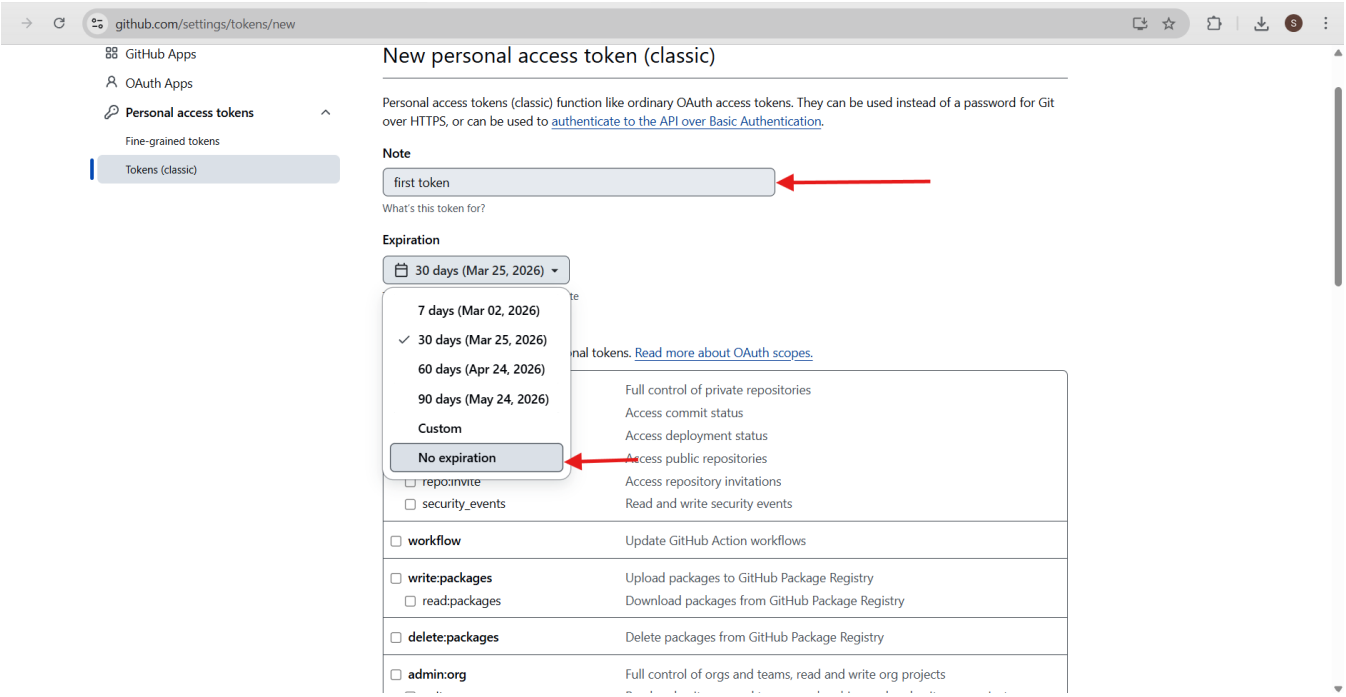
Step 11: after click on developer Settings, we need to select token(classic) from personal access token.



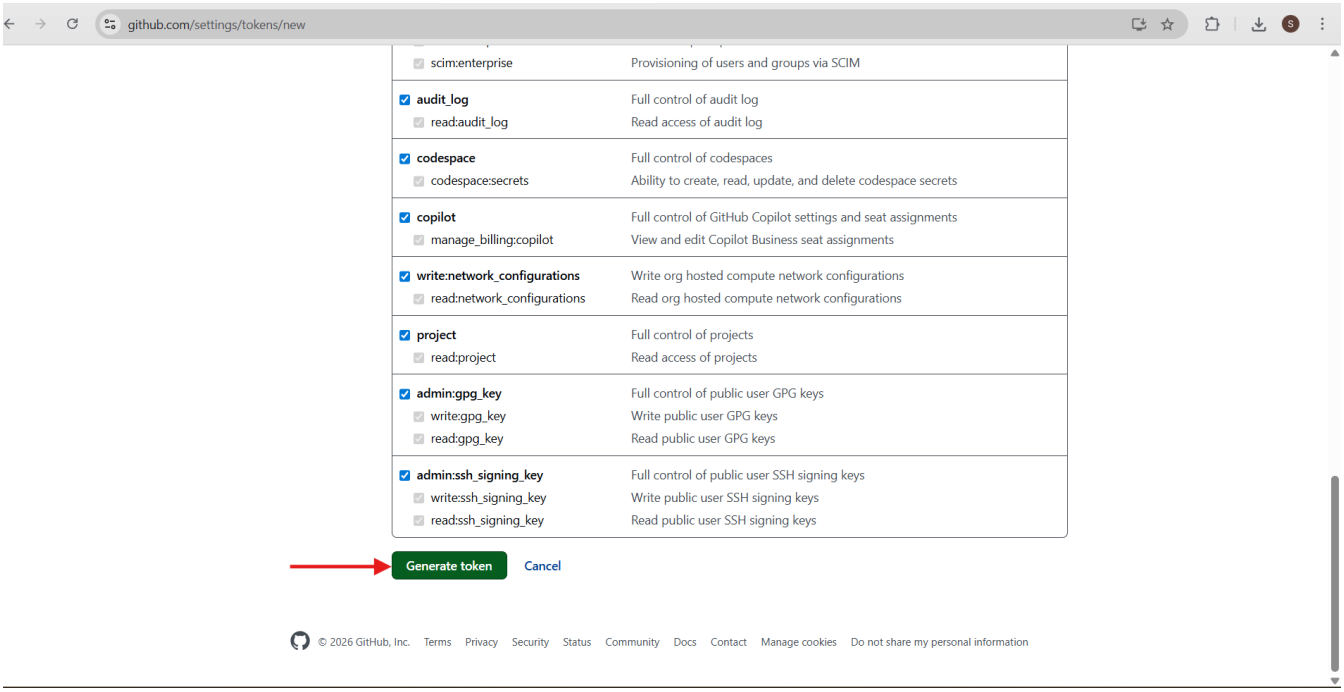
Step 12:Then we press on generate new token and choose (classic) one.



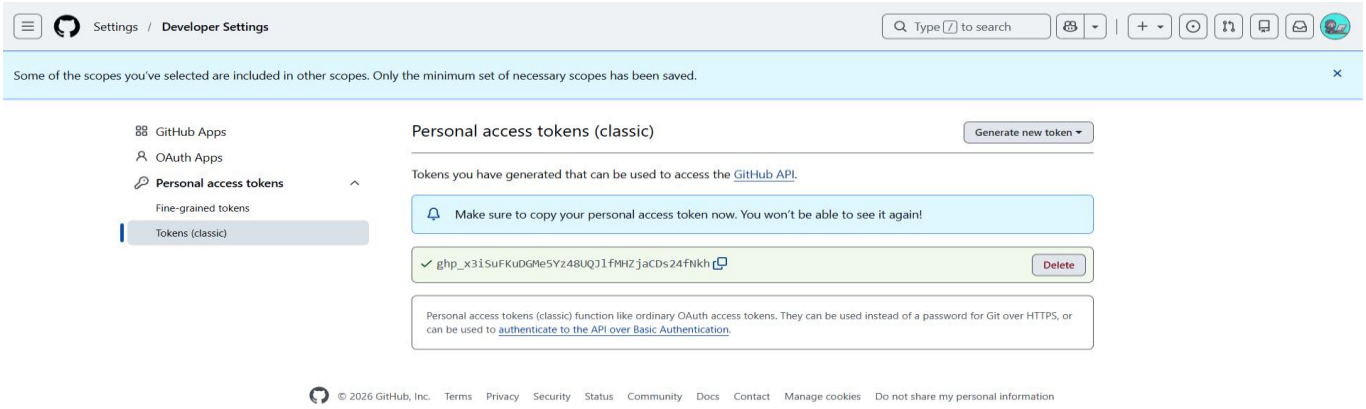
Step 13:then we give a note , and set expiration as “no expiration”. Checkin all the box.



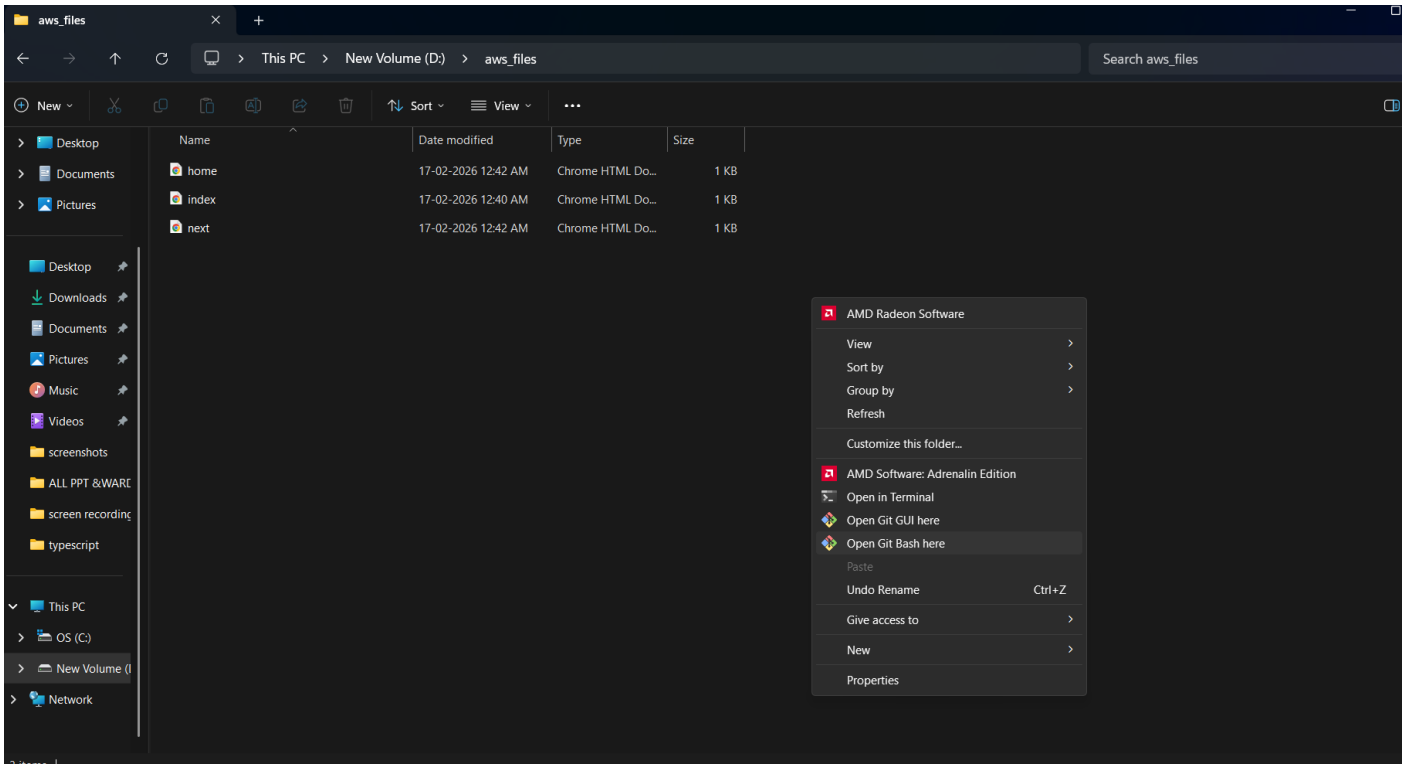
Step 14: Finally press on “ generate token”.



Step 15: my token is created.



Step 16: now we open the folder in my local machine where the project files are stored.
Step 17: after that, inside the folder we need to open git bash.



Step 18: in the git bash at first, we need delete all existing user, by running the command git config --global --unset-all user.name

```
sghar@Sanjib MINGW64 /d/aws_files
$ git config --global --unset-all user.name
```

Step 19: after that we need to setup our user's name and email by using specific command.

```
sghar@Sanjib MINGW64 /d/aws_files
$ git config --global user.name "sanjib994"
```

```
sghar@Sanjib MINGW64 /d/aws_files
$ git config --global user.email "sghara200@gmail.com"
```

Step 20: **next we need to check our setup.**

```
sghar@Sanjib MINGW64 /d/aws_files
$ git config --global --list
user.email=sghara200@gmail.com
user.name=sanjib994
```

Step 21: now initialize the file transfer from local machine to GitHub repositories by using git init command.

```
sghar@Sanjib MINGW64 /d/aws_files
$ git init
Initialized empty Git repository in D:/aws_files/.git/
```

Step 22: after that add files.

```
sghar@Sanjib MINGW64 /d/aws_files (master)
$ git add .
```

Step 23: for checking status we need to run git status command.

```
sghar@Sanjib MINGW64 /d/aws_files (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   home.html
        new file:   index.html
        new file:   next.html
```

Step 24: now we need to commit our files to the repo, for this we use “git commit -m “” command.

```
sghar@Sanjib MINGW64 /d/aws_files (master)
$ git commit -m "done"
[master (root-commit) 492d516] done
 3 files changed, 43 insertions(+)
 create mode 100644 home.html
 create mode 100644 index.html
 create mode 100644 next.html
```

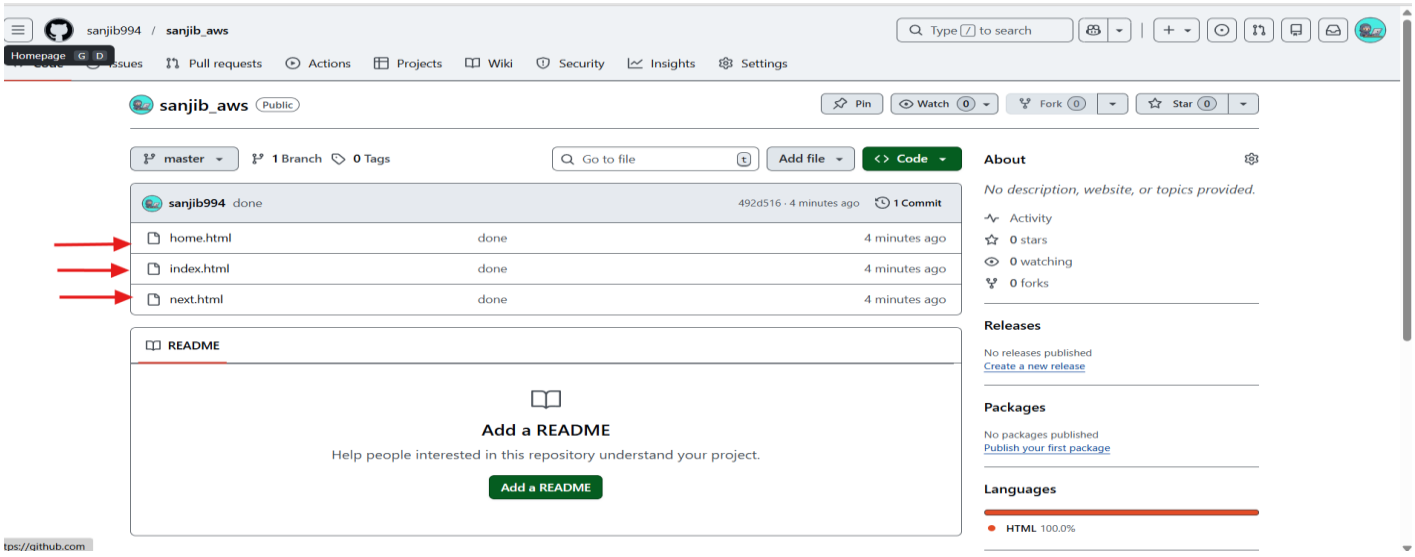
Step 25:now time to push the files into the destination repo by using two commands.

I. **First set the remote path**
II. **Then push the files to the destination path.**

```
sghar@Sanjib MINGW64 /d/aws_files (master)
$ git remote add origin https://github.com/sanjib994/sanjib_aws.git
```

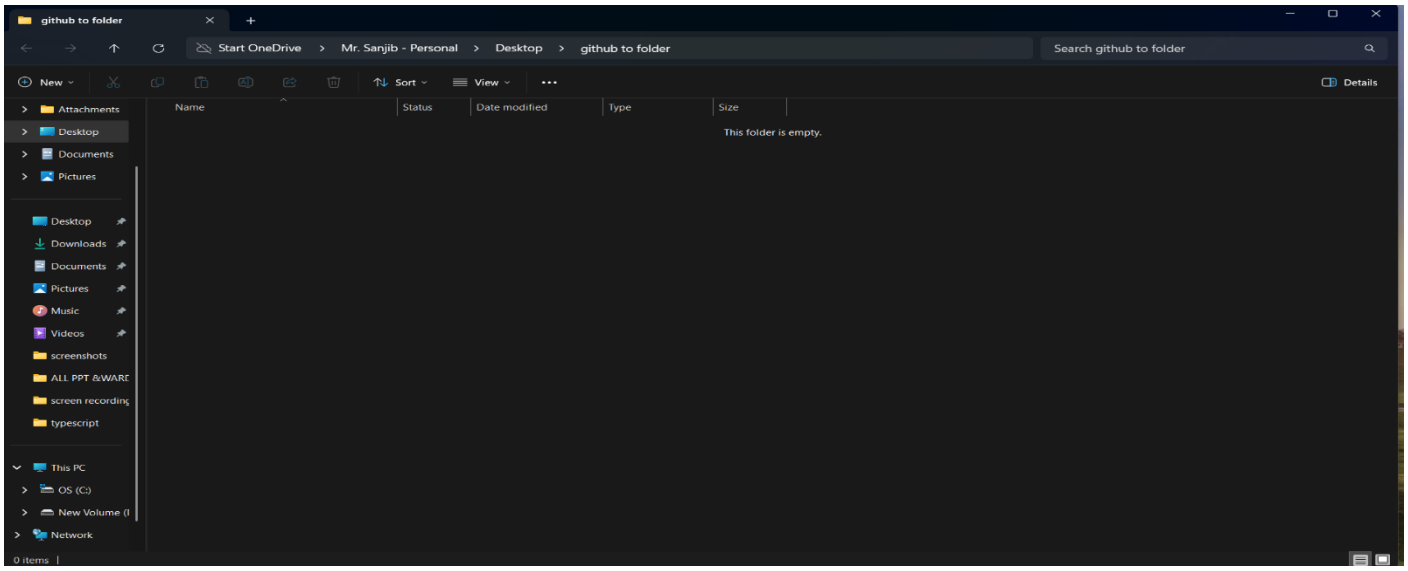
```
sghar@Sanjib MINGW64 /d/aws_files (master)
$ git push -u origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 606 bytes | 303.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/sanjib994/sanjib_aws.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

step 26: now we can see that the files are transfer successfully.

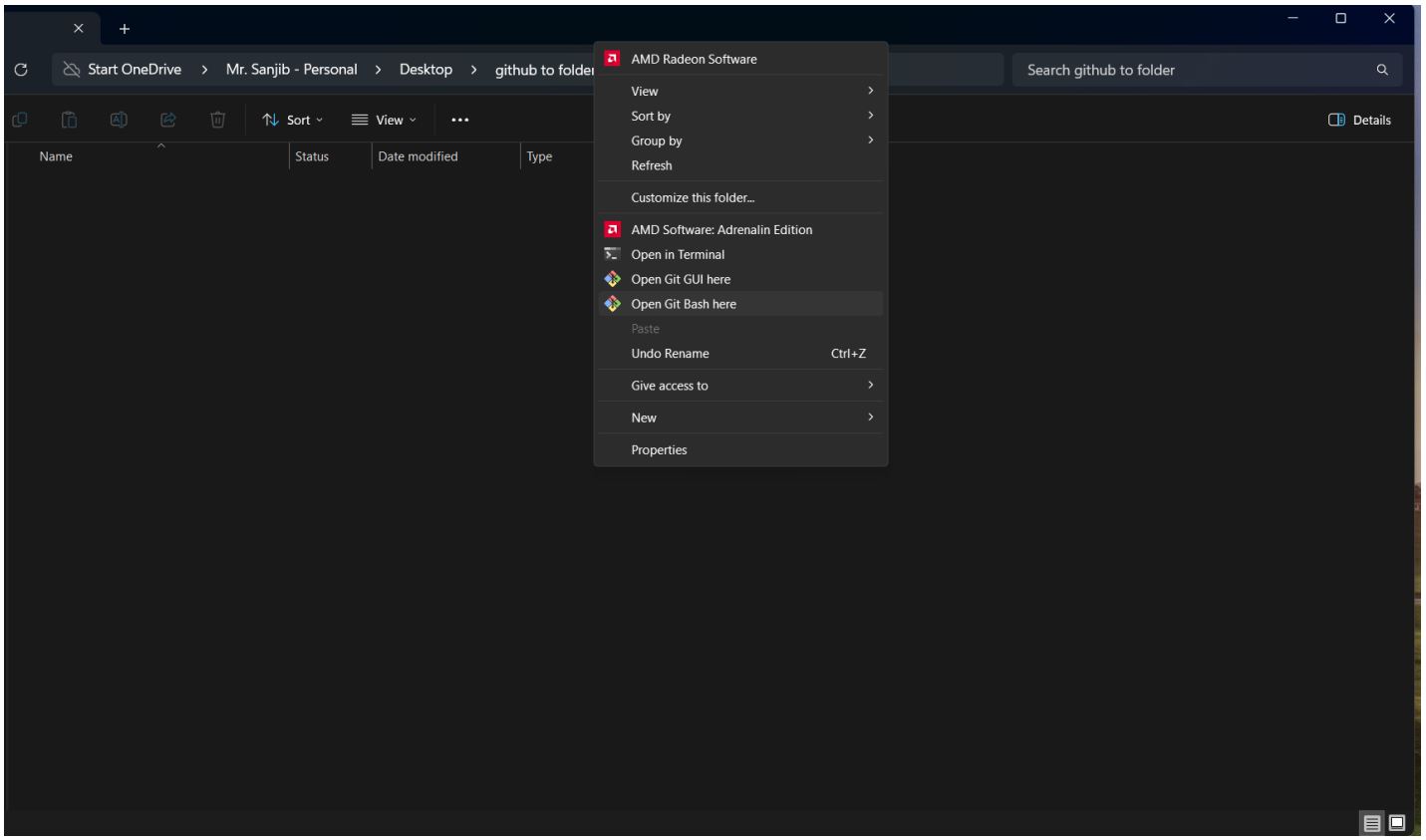


NOW WE DOTHE REVRESE THING MEANS COPY FILES FROM GITHUB TO LOCAL MECHINE.

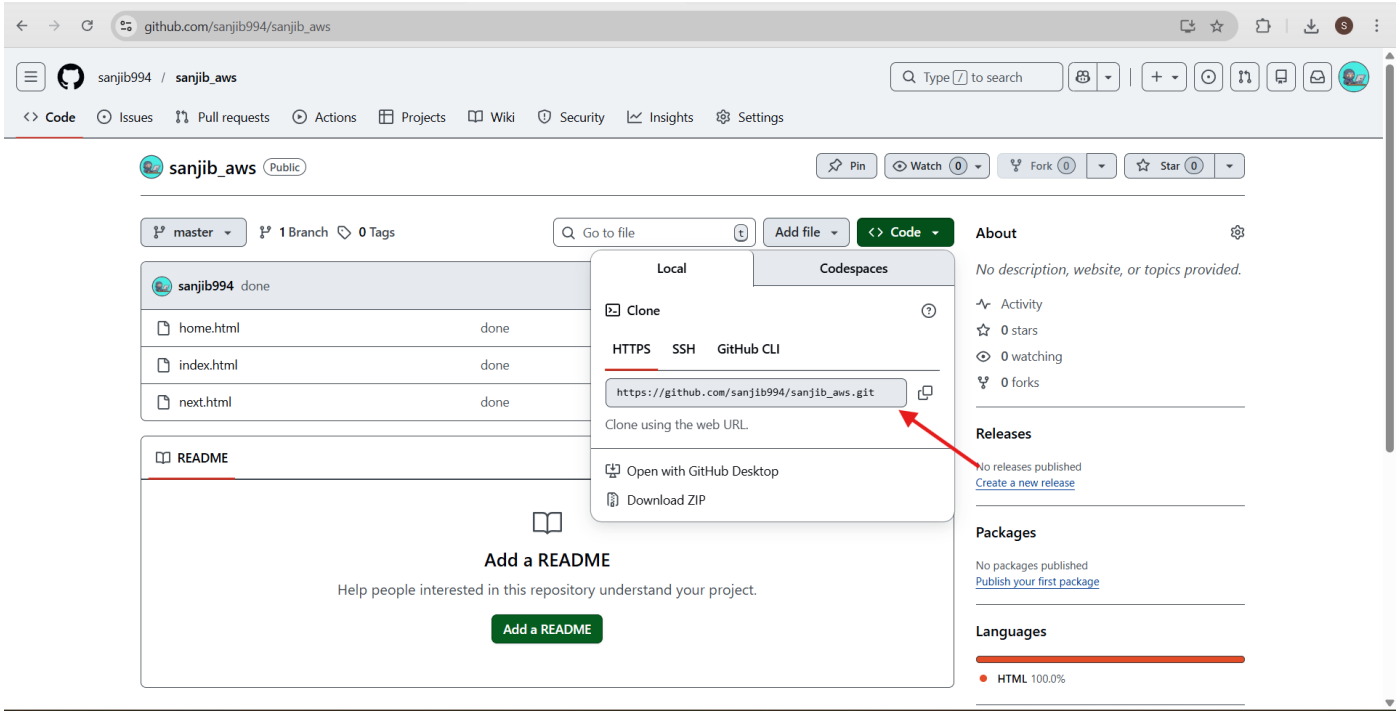
Step 1: create folder on local machine desktop.



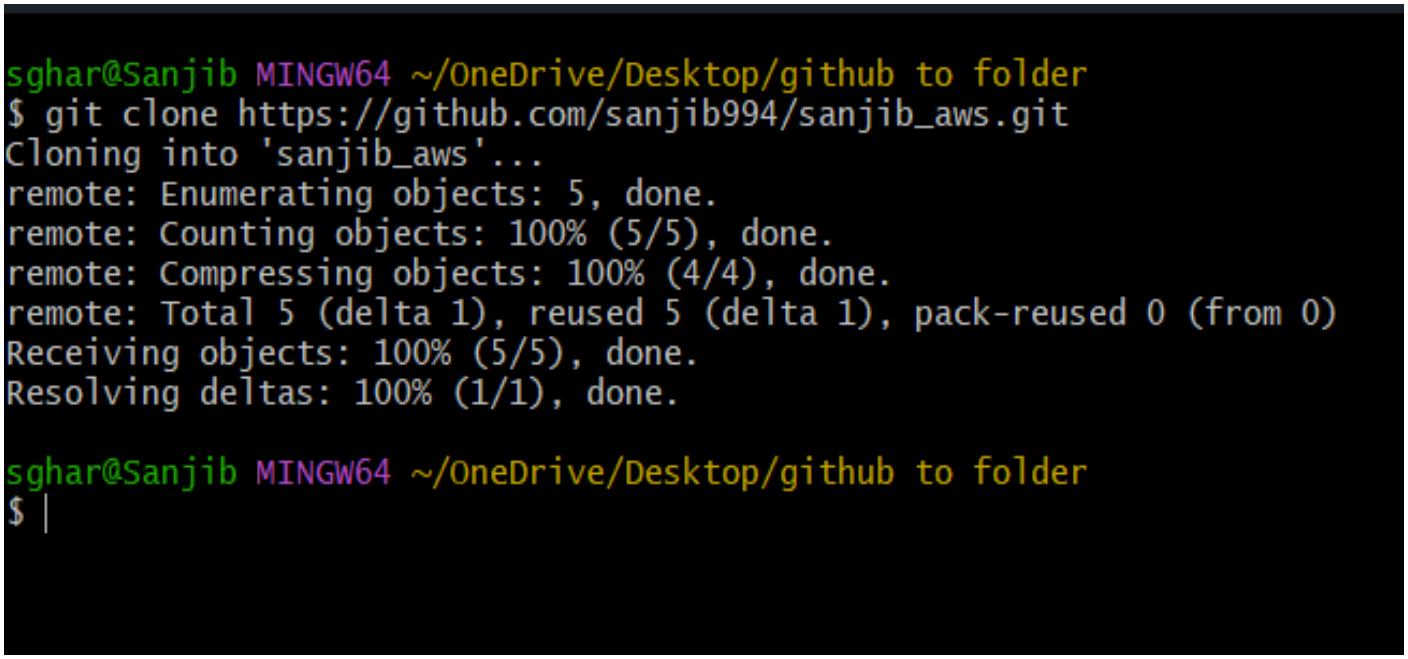
Step 2: then open folder. Also open the git bash app inside the folder.



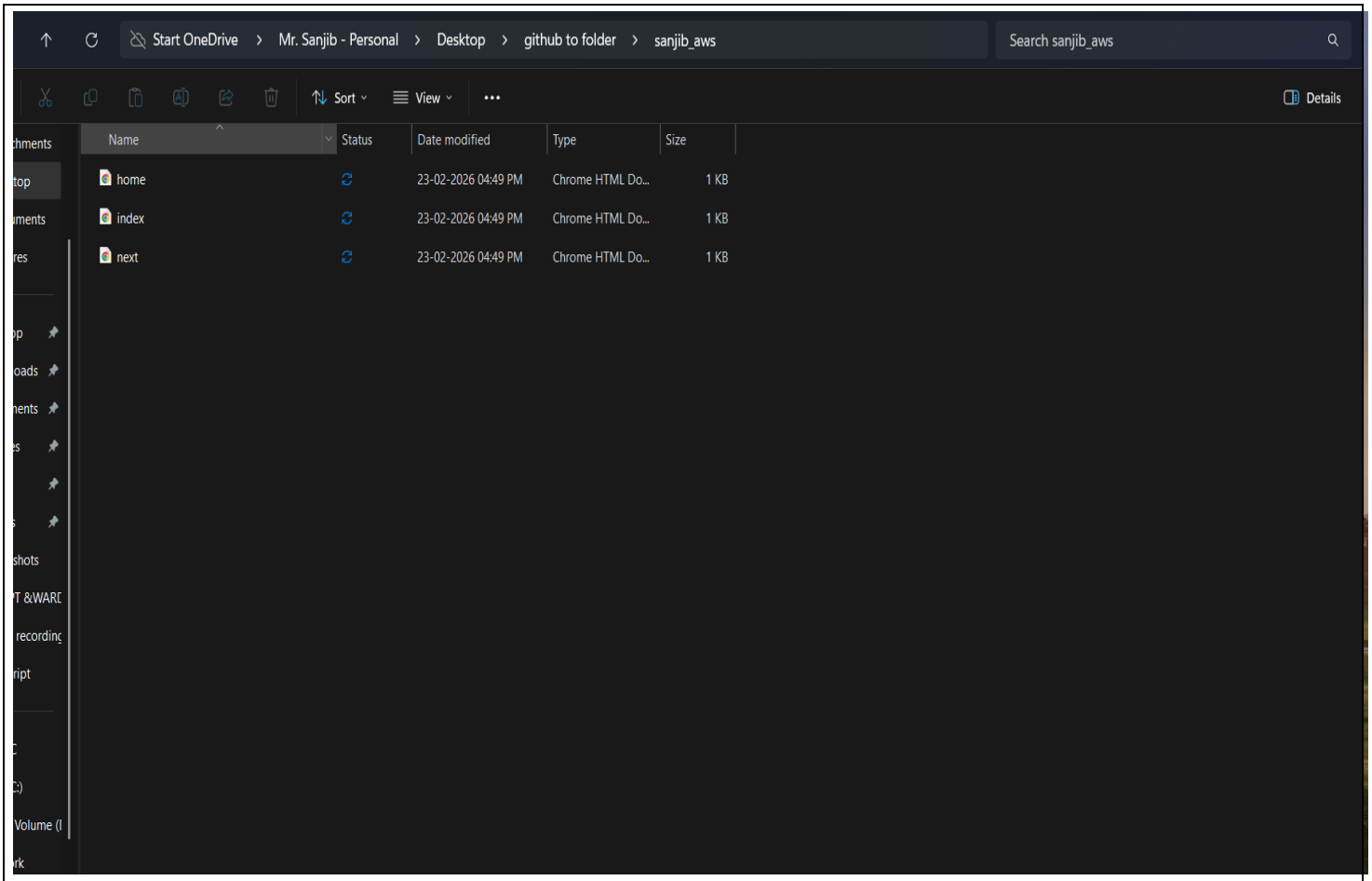
Step 3: after that we need to copy the URL from GitHub repo.

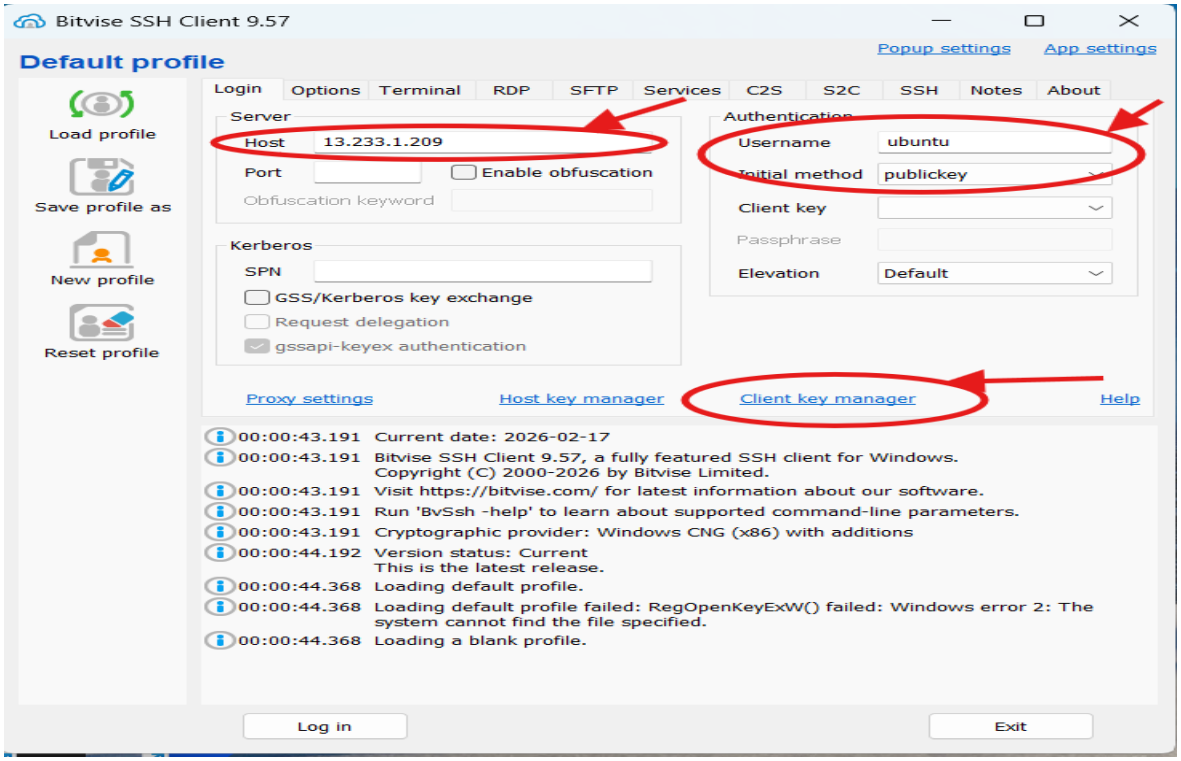


Step 4: then run a command in the git bash to copy the files from GitHub repo to local machine.



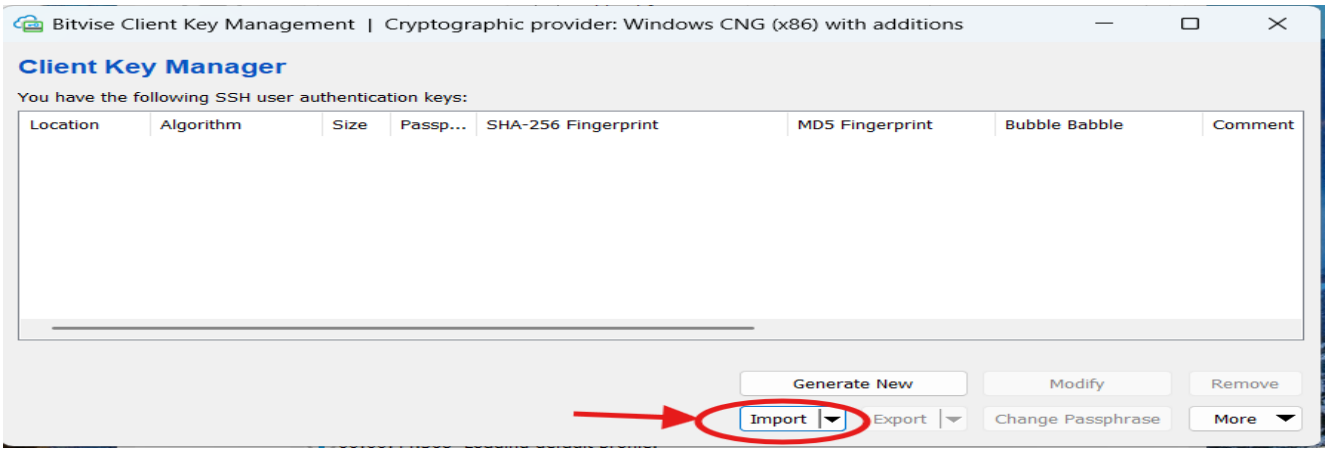
Step 5: folder transfer successfully.

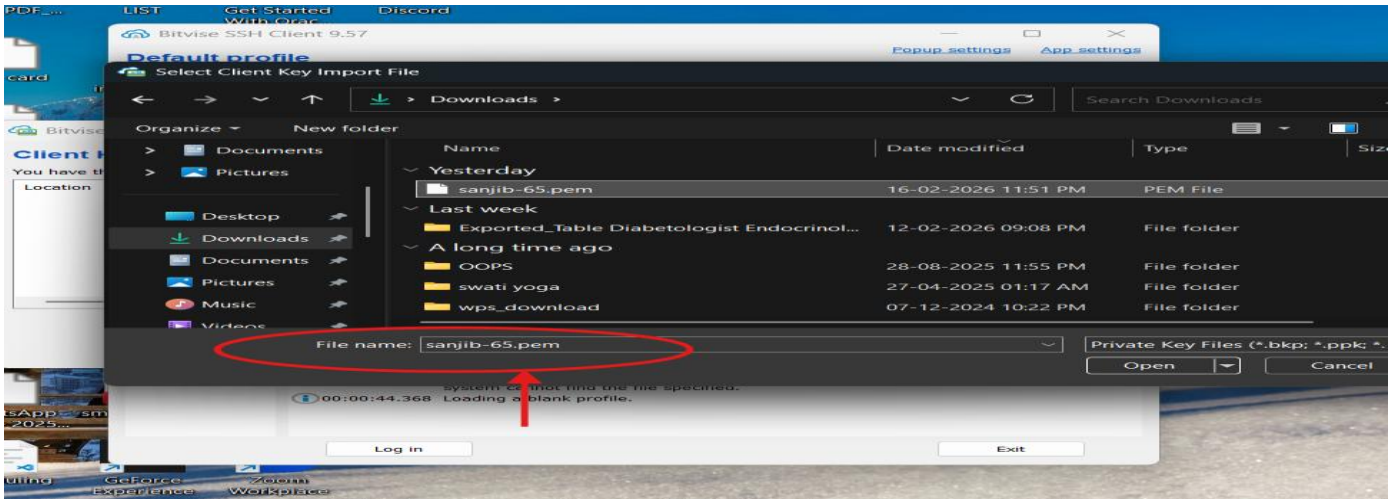

Start OneDrive
Mr. Sanjib - Personal
Desktop
github to folder
sanjib_aws
Search sanjib_aws
Sort
View
Details
Name
Status
Date modified
Type
Size
home
23-02-2026 04:49 PM
Chrome HTML Do...
1 KB
index
23-02-2026 04:49 PM
Chrome HTML Do...
1 KB
next
23-02-2026 04:49 PM
Chrome HTML Do...
1 KB

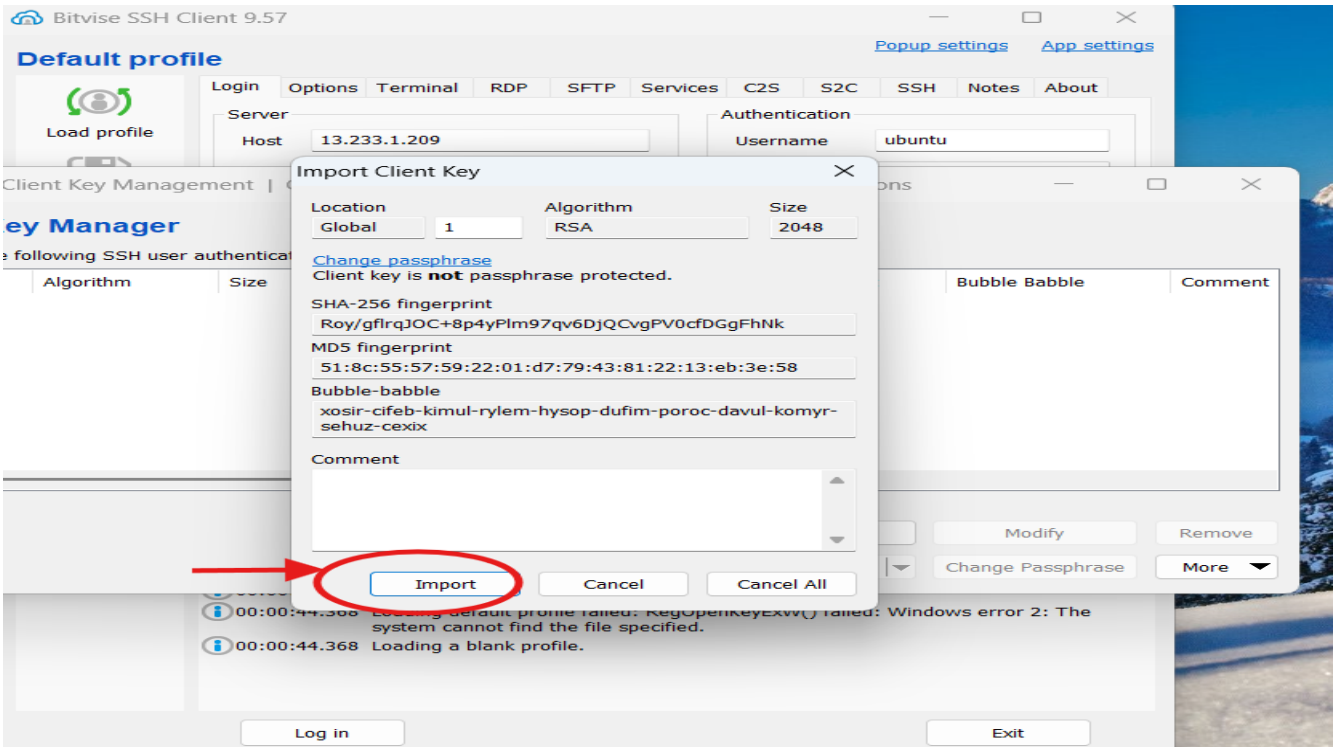


Step 4: Import Private Key

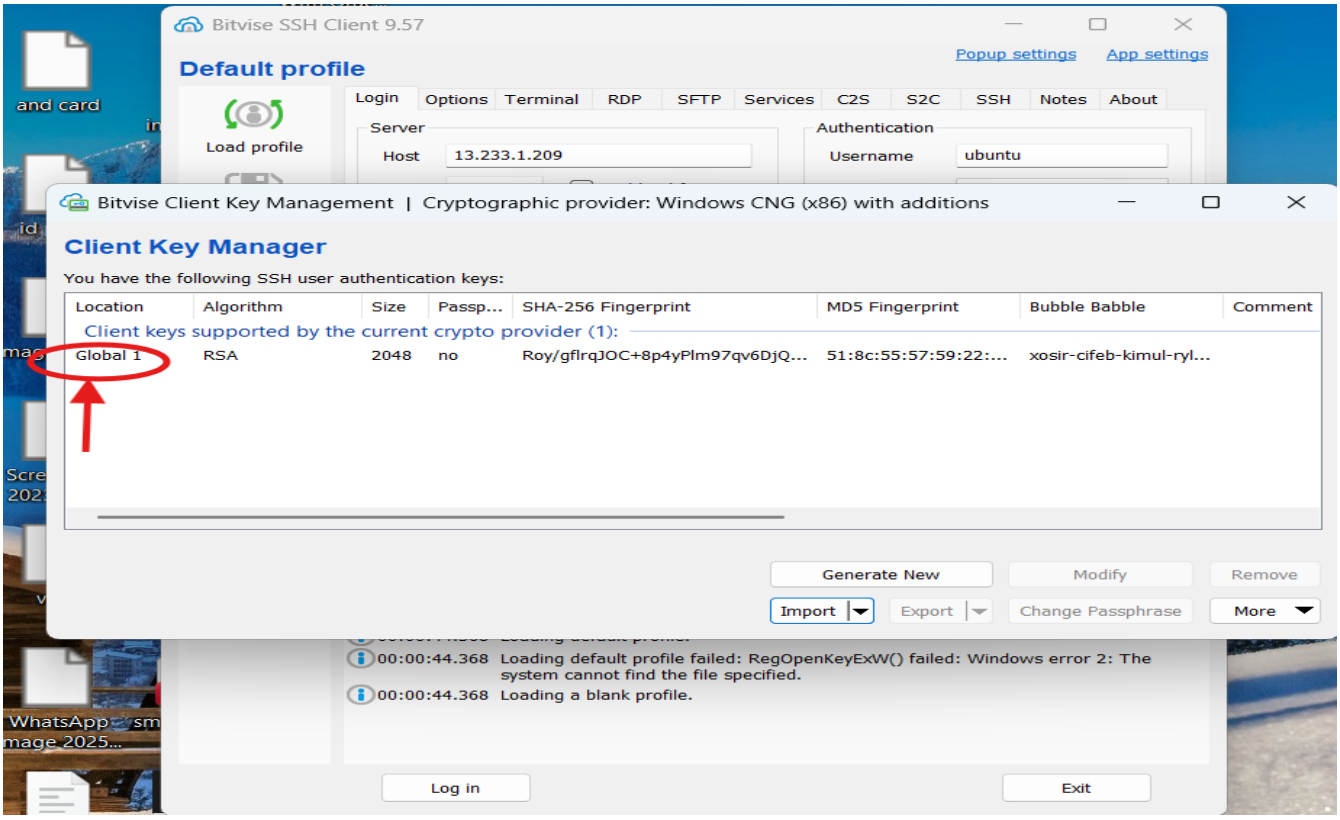
1. In Client Key Manager, click Import.
2. Select the downloaded .pem key file from the Desktop.





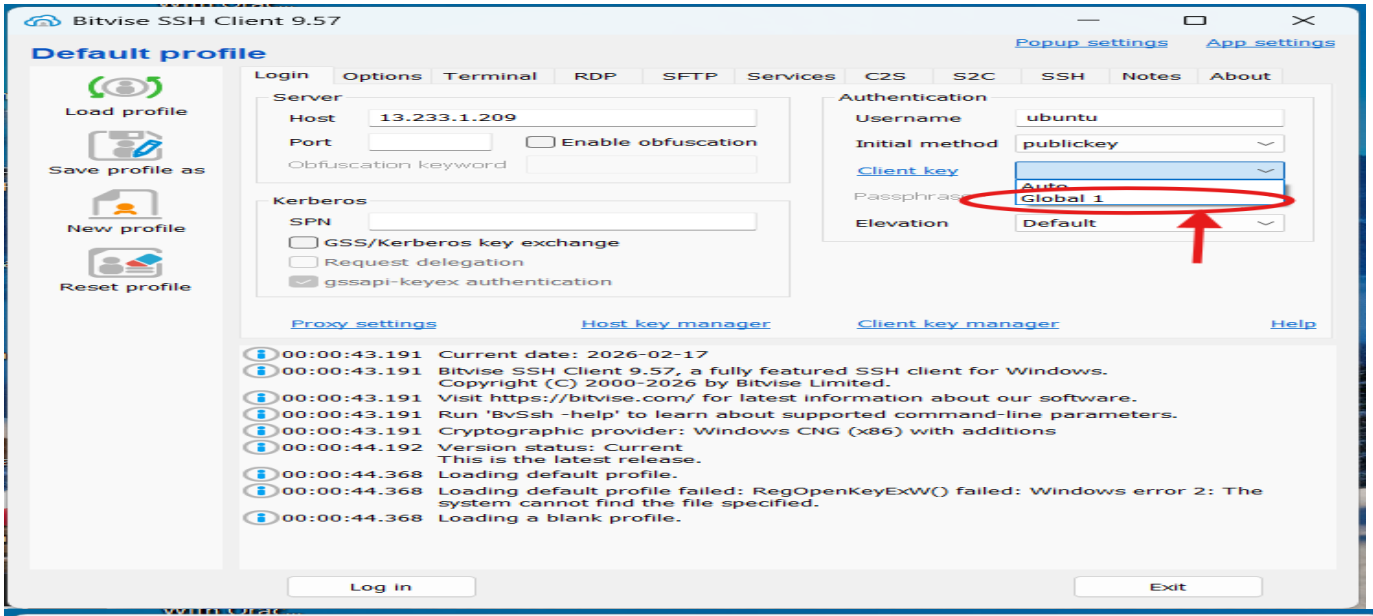


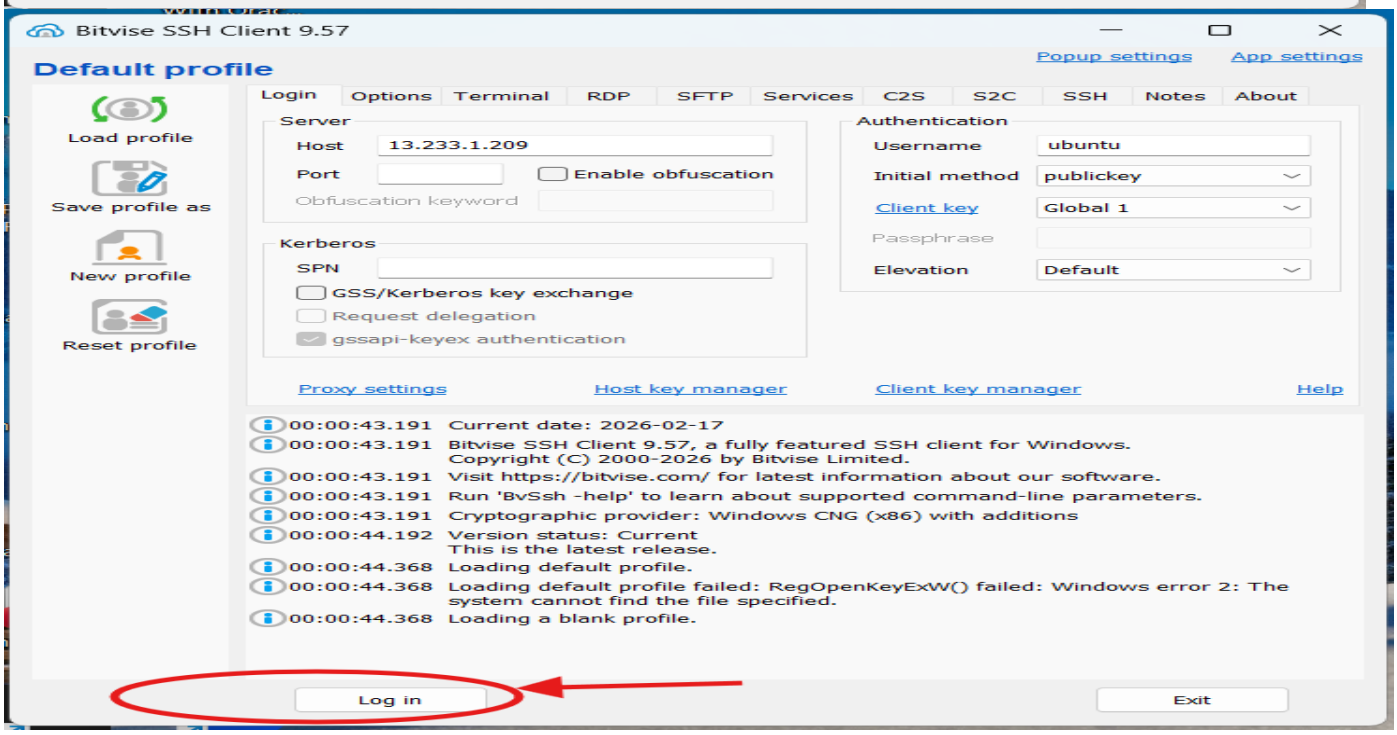
3.Save the key.



Step 5: Login to the Instance

1. Ensure Client key is set to Global 1.
2. Click on the Log in button.

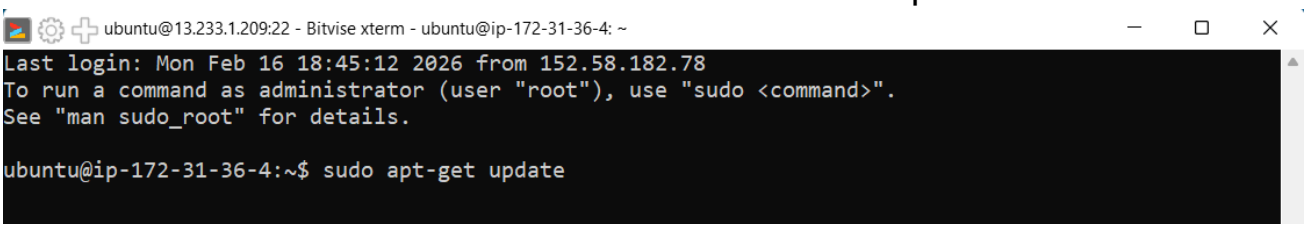




Step 6: Open Terminal Console

After successful login:

- Click on **New Terminal Console** from the **Bitvise** side panel.



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Step 7: Update Ubuntu System

Run the following command to update and upgrade the system:

```
Get:56 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [396 B]
Fetched 39.9 MB in 6s (6529 kB/s)
Reading package lists... Done
ubuntu@ip-172-31-36-4:~$ sudo apt-get upgrade
```

sudo apt-get update && sudo apt-get upgrade -y

Step 8: Install Nginx Web Server

Install Nginx using the command:

```
User sessions running outdated binaries:
 ubuntu @ session #2: sshd[1061,1172]
 ubuntu @ user manager service: systemd[1066]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-36-4:~$ sudo apt-get install nginx
```

sudo apt-get install nginx -y

```
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 565 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

Step 9: Set Permissions for Web Directory

Navigate to the web directory:

cd /var/www

```
ubuntu@ip-172-31-36-4:/home$ cd..
cd..: command not found
ubuntu@ip-172-31-36-4:/home$ cd ..
ubuntu@ip-172-31-36-4:/$ ls
bin                boot  etc   lib               lib64       media  opt   root  sbin               snap  sys  usr
bin.usr-is-merged  dev   home  lib.usr-is-merged lost+found  mnt    proc  run   sbin.usr-is-merged srv   tmp  var
ubuntu@ip-172-31-36-4:/$
```

```
ubuntu@ip-172-31-36-4:/$ cd var
ubuntu@ip-172-31-36-4:/var$
```

```
ubuntu@ip-172-31-36-4:/var$ ls
backups  cache  crash  lib  local  lock  log  mail  opt  run  snap  spool  tmp  www
ubuntu@ip-172-31-36-4:/var$
```

```
ubuntu@ip-172-31-36-4:/var$ cd www/
ubuntu@ip-172-31-36-4:/var/www$ ls
html
ubuntu@ip-172-31-36-4:/var/www$
```

Give proper permissions to the HTML directory:

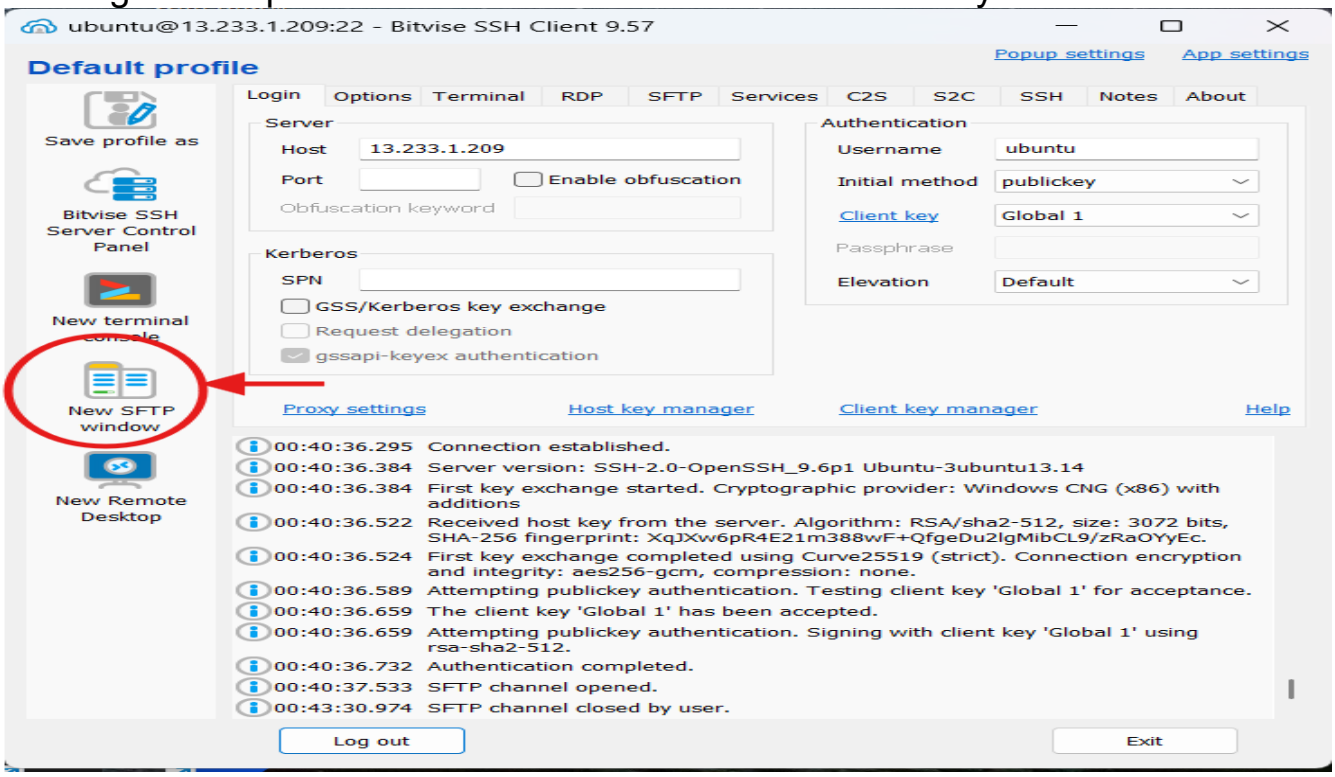
sudo chmod 777 html/

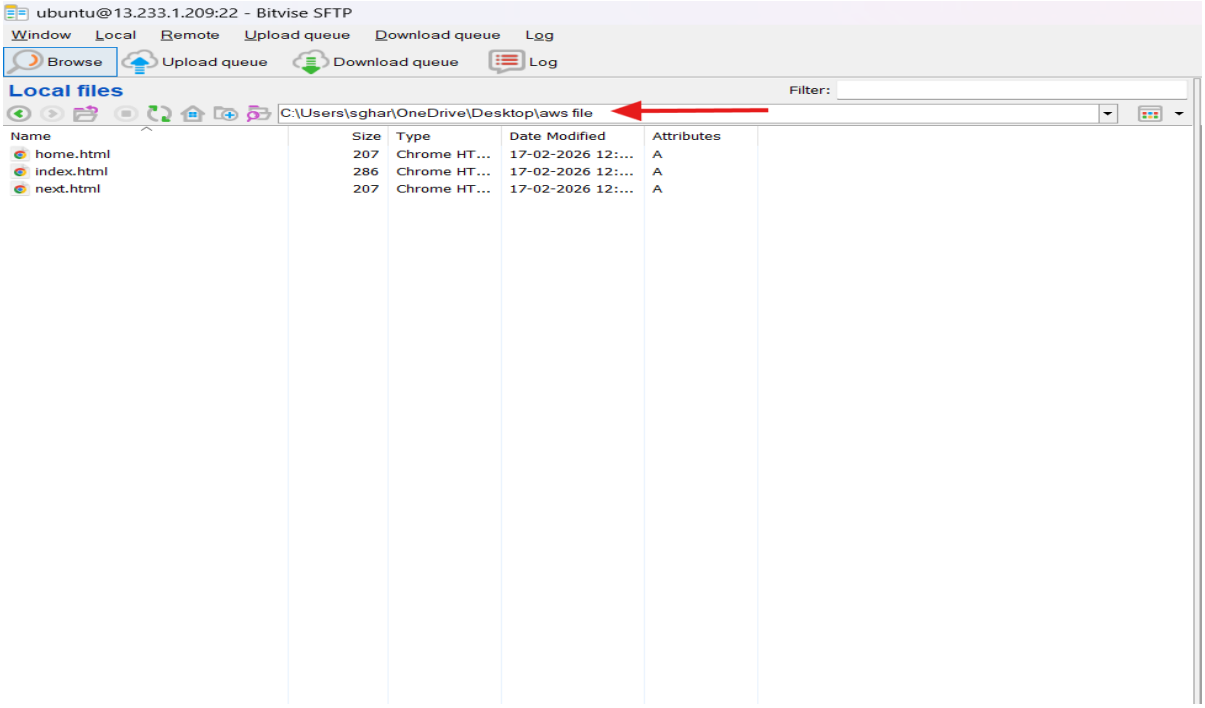
```
ubuntu@ip-172-31-36-4:/var/www$ ls
html
ubuntu@ip-172-31-36-4:/var/www$ sudo chmod 777 html
ubuntu@ip-172-31-36-4:/var/www$
```

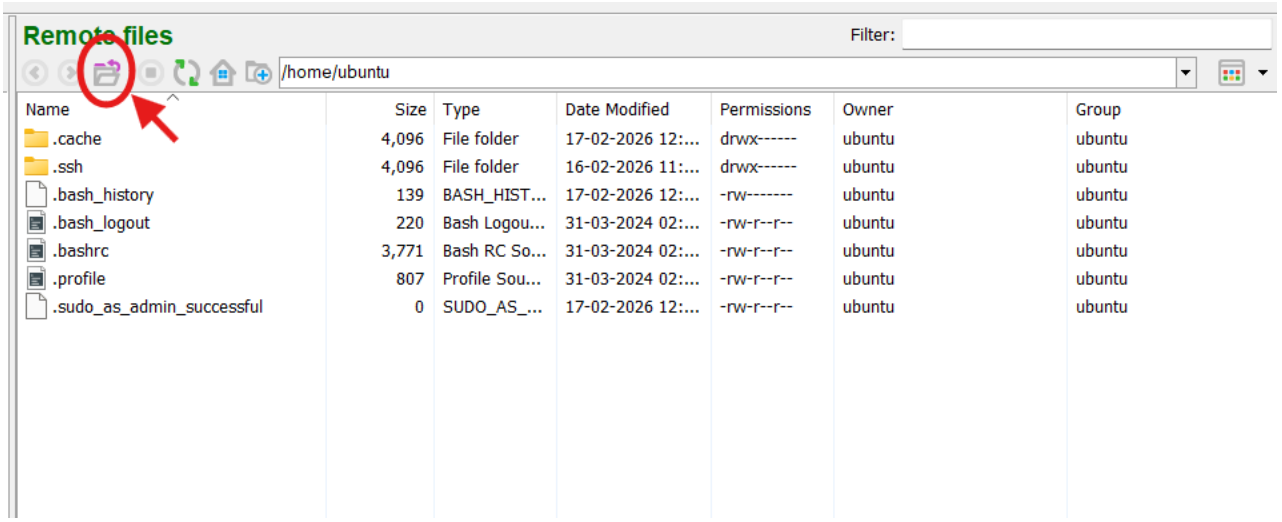
Step 10: Upload Website Files Using SFTP

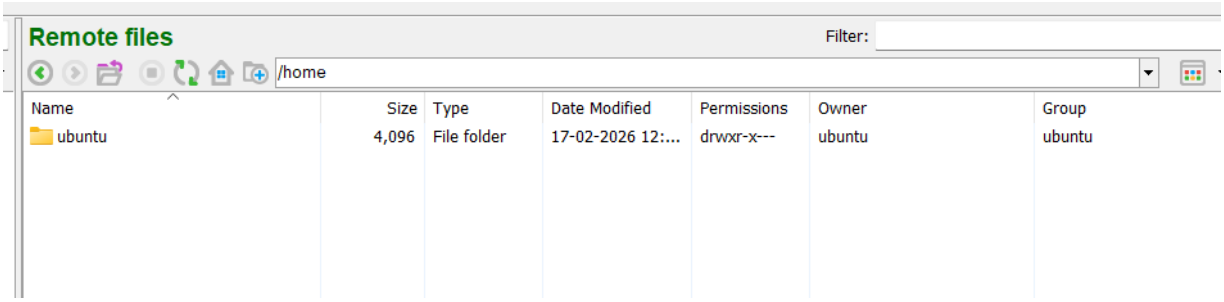
1. Click New SFTP window in Bitvise.
2. In the left pane (Local Files), navigate to the folder containing:
    - index.html
    - prev.html
    - next.html

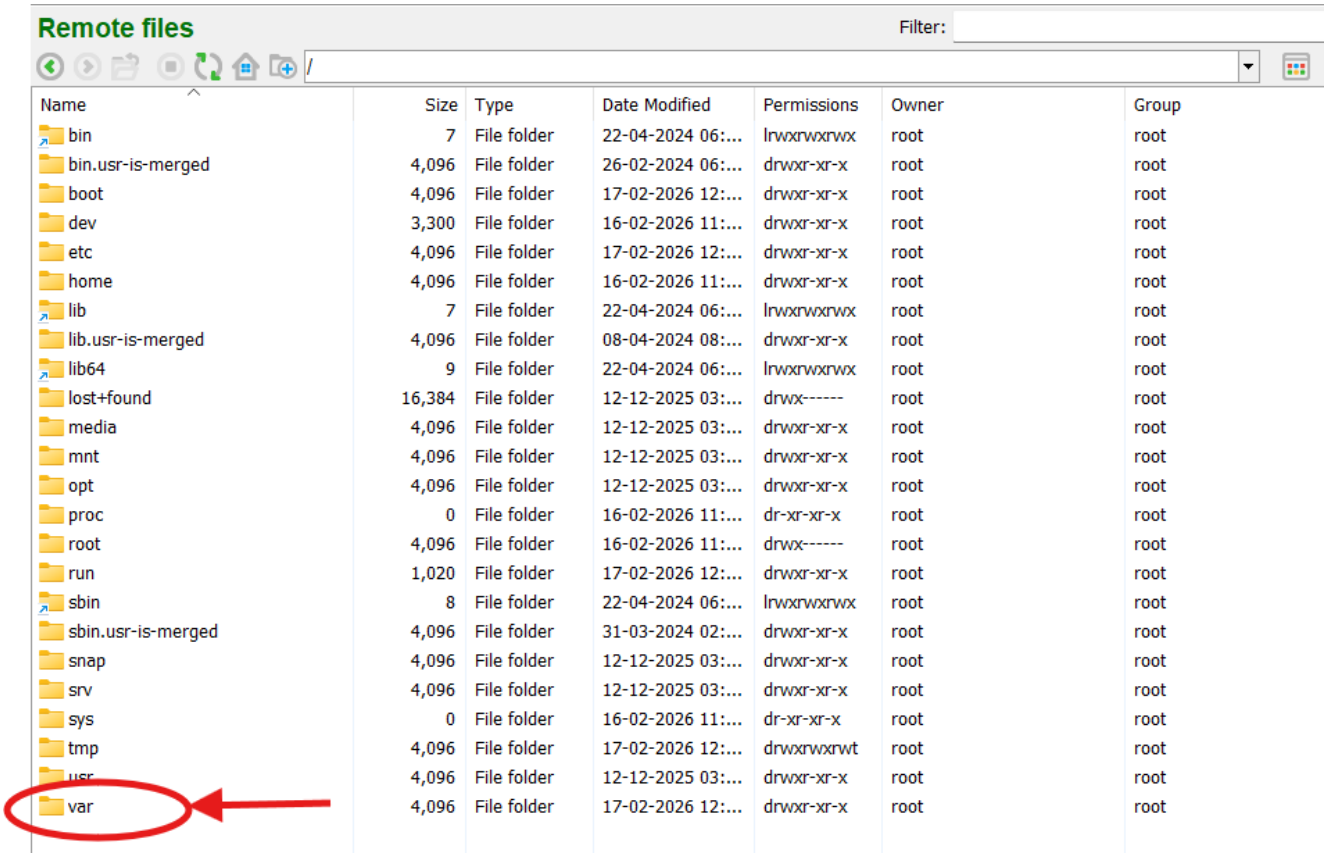
3. In the right pane (Remote Files), navigate to: **/var/www/html**
4. Drag and drop all three files from local to remote directory.

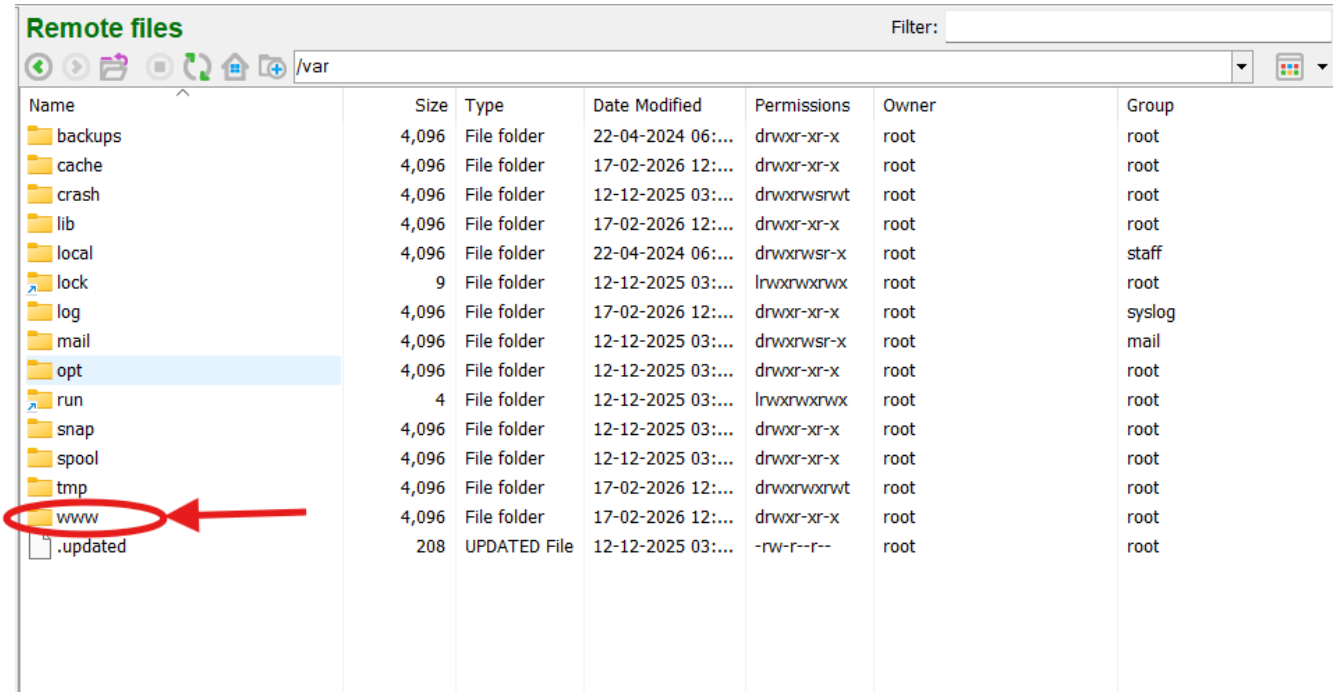


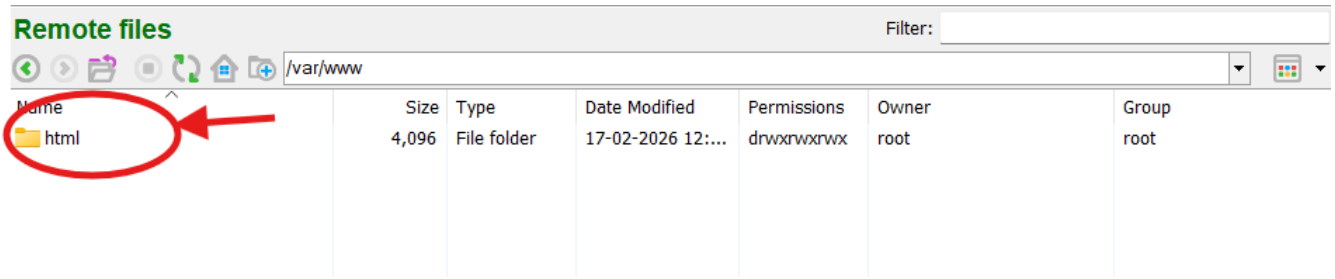


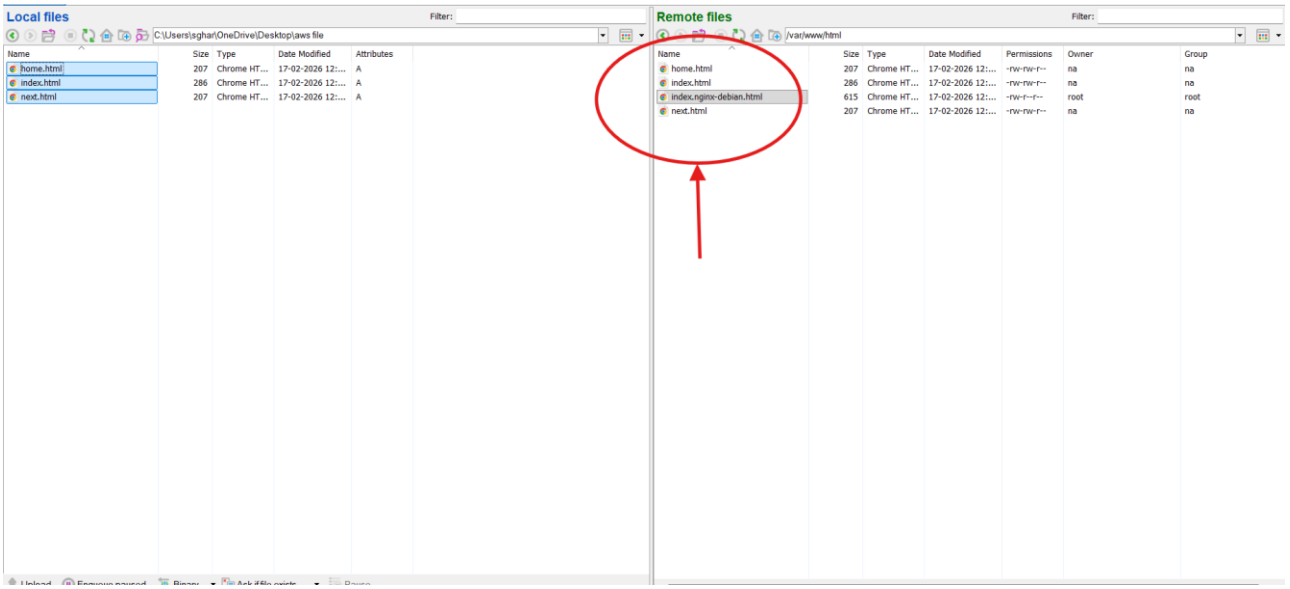

Remote files
Filter:
/home/ubuntu
Name Size Type Date Modified Permissions Owner Group
.cache 4,096 File folder 17-02-2026 12:... drwx------ ubuntu ubuntu
.ssh 4,096 File folder 16-02-2026 11:... drwx------ ubuntu ubuntu
.bash_history 139 BASH_HIST... 17-02-2026 12:... -rw------- ubuntu ubuntu
.bash_logout 220 Bash Logou... 31-03-2024 02:... -rw-r--r-- ubuntu ubuntu
.bashrc 3,771 Bash RC So... 31-03-2024 02:... -rw-r--r-- ubuntu ubuntu
.profile 807 Profile Sou... 31-03-2024 02:... -rw-r--r-- ubuntu ubuntu
.sudo_as_admin_successful 0 SUDO_AS_... 17-02-2026 12:... -rw-r--r-- ubuntu ubuntu


Remote files
Filter:
/home
Name Size Type Date Modified Permissions Owner Group
ubuntu 4,096 File folder 17-02-2026 12:... drwxr-x--- ubuntu ubuntu


Remote files
Filter:
/
Name Size Type Date Modified Permissions Owner Group
bin 7 File folder 22-04-2024 06:... lrwxrwxrwx root root
bin.usr-is-merged 4,096 File folder 26-02-2024 06:... drwxr-xr-x root root
boot 4,096 File folder 17-02-2026 12:... drwxr-xr-x root root
dev 3,300 File folder 16-02-2026 11:... drwxr-xr-x root root
etc 4,096 File folder 17-02-2026 12:... drwxr-xr-x root root
home 4,096 File folder 16-02-2026 11:... drwxr-xr-x root root
lib 7 File folder 22-04-2024 06:... lrwxrwxrwx root root
lib.usr-is-merged 4,096 File folder 08-04-2024 08:... drwxr-xr-x root root
lib64 9 File folder 22-04-2024 06:... lrwxrwxrwx root root
lost+found 16,384 File folder 12-12-2025 03:... drwx------ root root
media 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
mnt 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
opt 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
proc 0 File folder 16-02-2026 11:... dr-xr-xr-x root root
root 4,096 File folder 16-02-2026 11:... drwx------ root root
run 1,020 File folder 17-02-2026 12:... drwxr-xr-x root root
sbin 8 File folder 22-04-2024 06:... lrwxrwxrwx root root
sbin.usr-is-merged 4,096 File folder 31-03-2024 02:... drwxr-xr-x root root
snap 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
srv 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
sys 0 File folder 16-02-2026 11:... dr-xr-xr-x root root
tmp 4,096 File folder 17-02-2026 12:... drwxrwxrwt root root
usr 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
var 4,096 File folder 17-02-2026 12:... drwxr-xr-x root root


Remote files
Filter:
/var
Name Size Type Date Modified Permissions Owner Group
backups 4,096 File folder 22-04-2024 06:... drwxr-xr-x root root
cache 4,096 File folder 17-02-2026 12:... drwxr-xr-x root root
crash 4,096 File folder 12-12-2025 03:... drwxrwsrwt root root
lib 4,096 File folder 17-02-2026 12:... drwxr-xr-x root root
local 4,096 File folder 22-04-2024 06:... drwxrwsr-x root staff
lock 9 File folder 12-12-2025 03:... lrwxrwxrwx root root
log 4,096 File folder 17-02-2026 12:... drwxr-xr-x root syslog
mail 4,096 File folder 12-12-2025 03:... drwxrwsr-x root mail
opt 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
run 4 File folder 12-12-2025 03:... lrwxrwxrwx root root
snap 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
spool 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
tmp 4,096 File folder 17-02-2026 12:... drwxrwxrwt root root
www 4,096 File folder 17-02-2026 12:... drwxr-xr-x root root
.updated 208 UPDATED File 12-12-2025 03:... -rw-r--r-- root root


Remote files
Filter:
/var/www
Name Size Type Date Modified Permissions Owner Group
html 4,096 File folder 17-02-2026 12:... drwxrwxrwx root root


Local files
Remote files
home.html
index.html
index.nginx-debian.html
next.html

Step 11: Access the Website

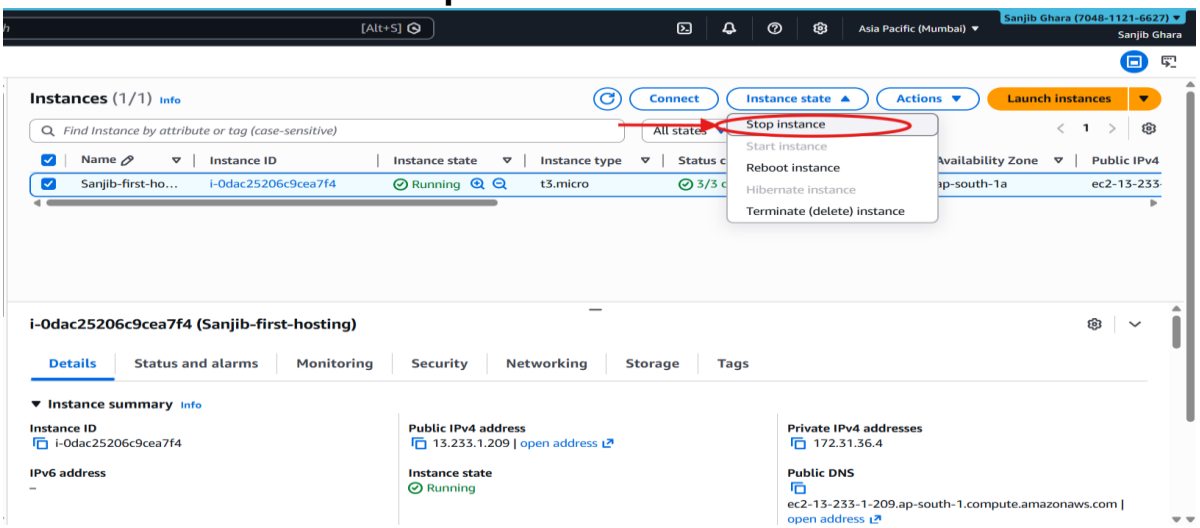
Open a new browser tab and enter:

http://<Public-IPv4-address>

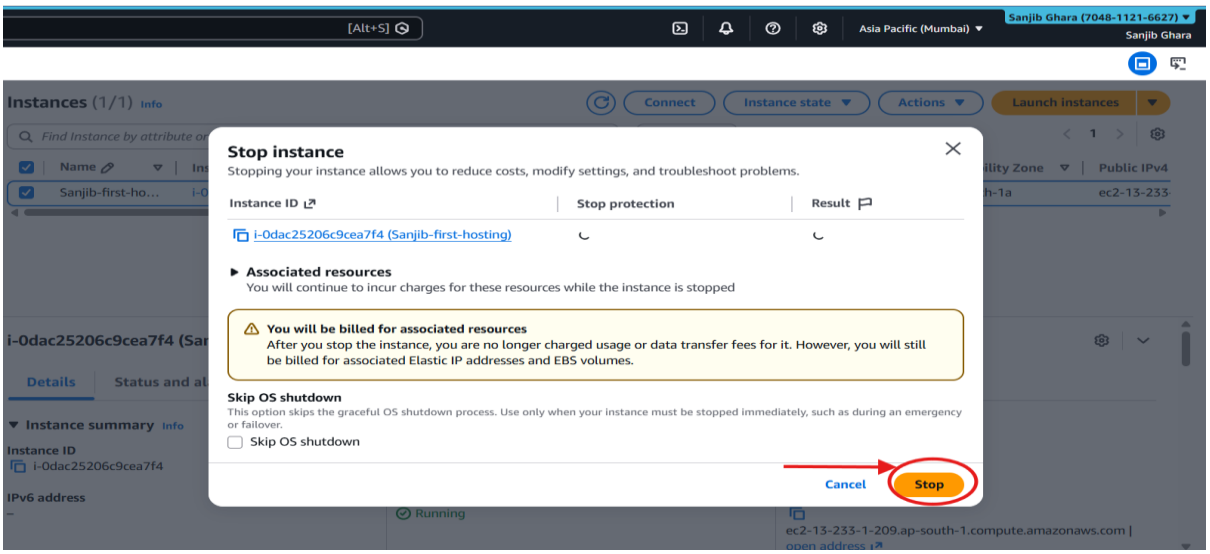
The static website will load successfully.

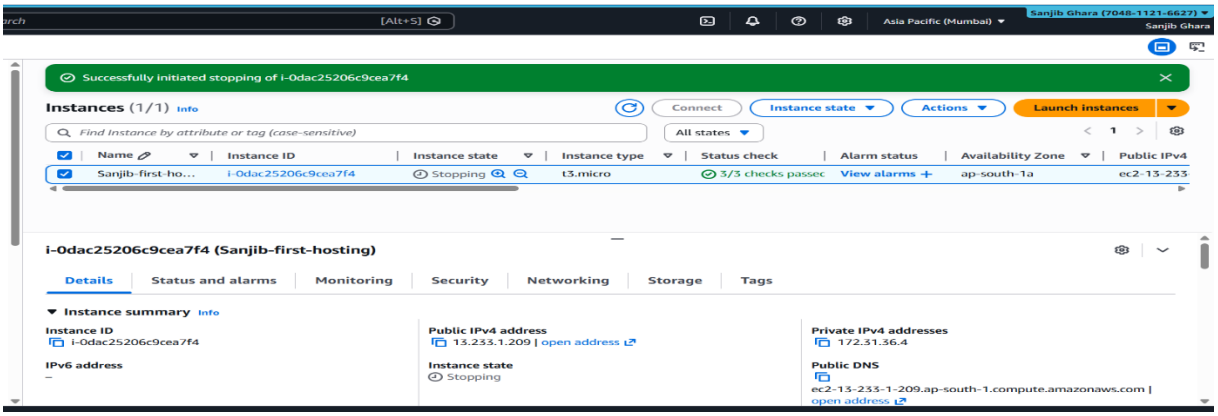
Step 12: Stop the EC2 Instance

1. Go back to the EC2 Instances page.
2. Select the instance.
3. Click Instance state → **Stop instance.**



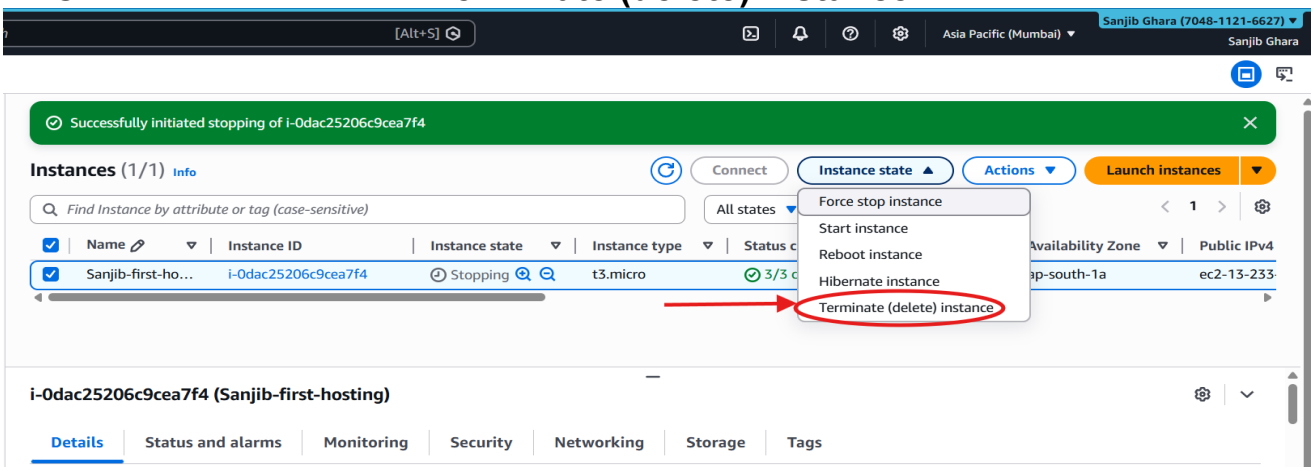
4. Confirm by clicking Stop.



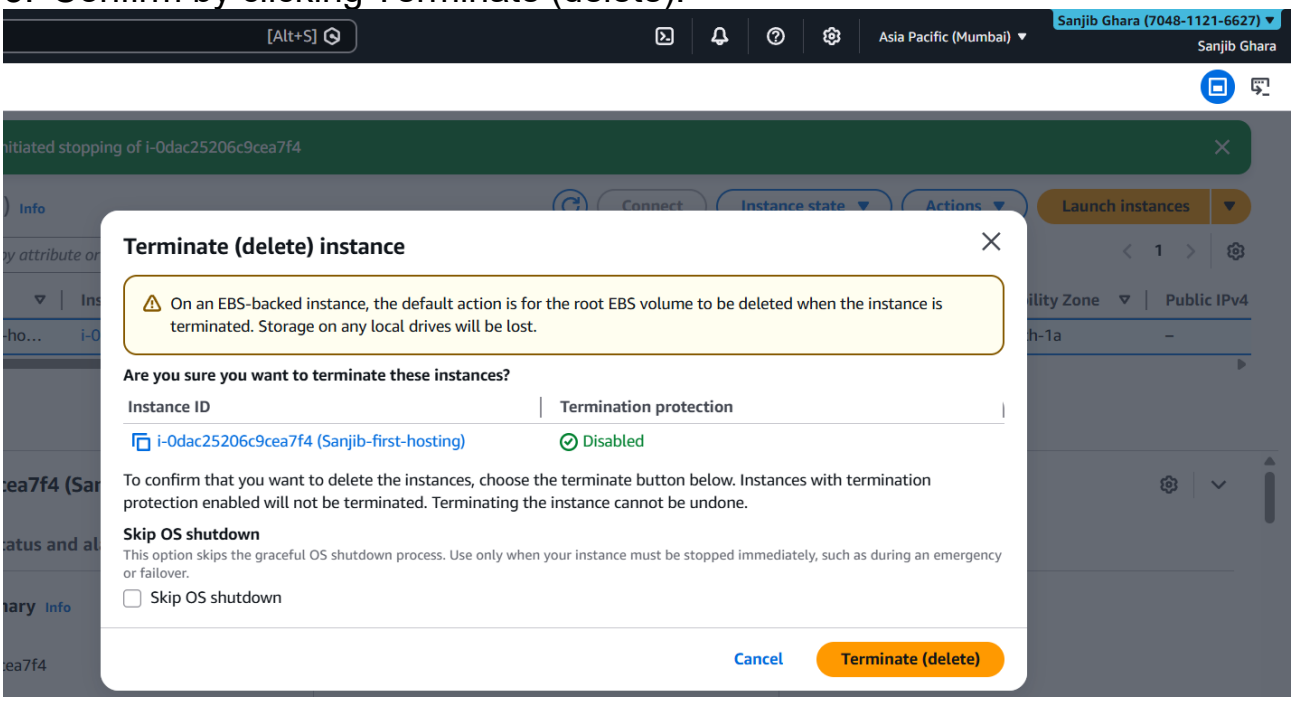


Step 13: Terminate the EC2 Instance

1. Select the same instance again.
2. Click Instance state → **Terminate (delete) instance.**

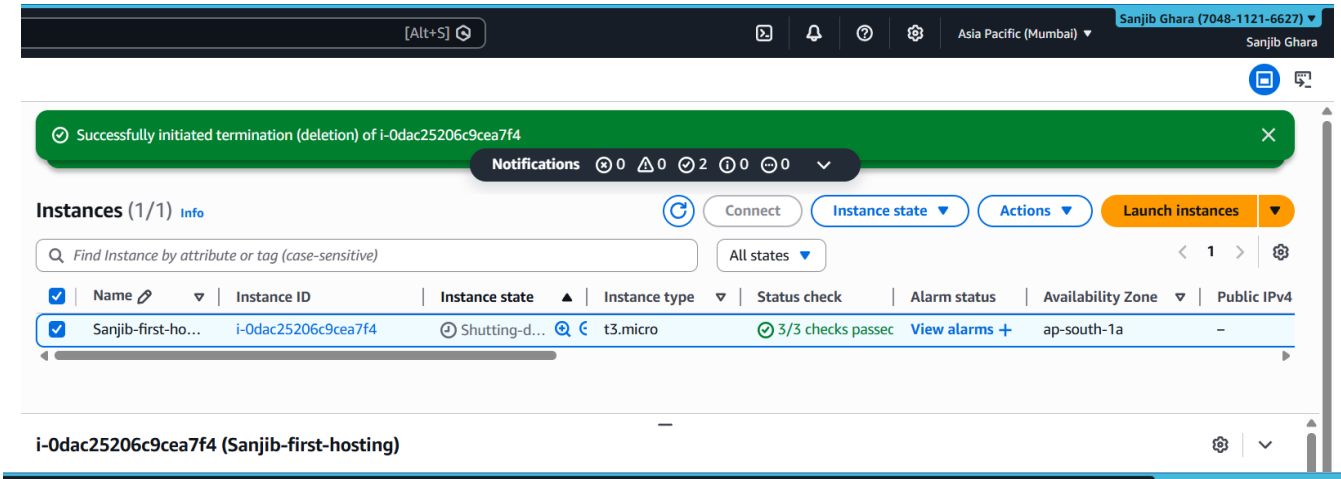


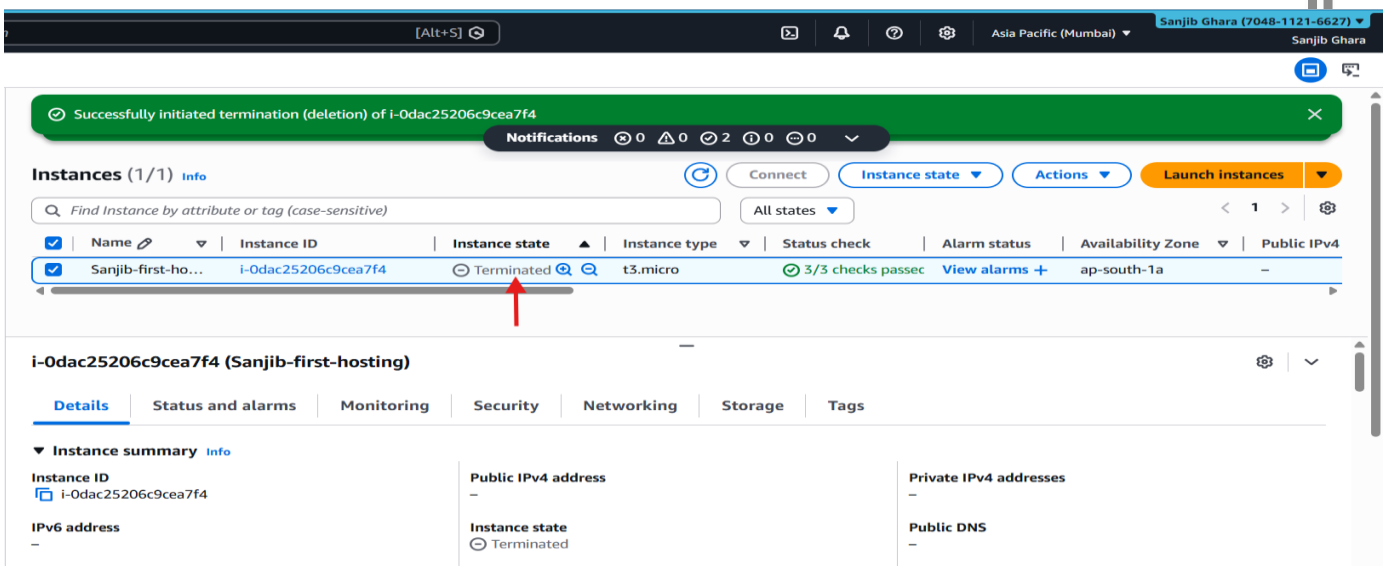
3. Confirm by clicking Terminate (delete).



Step 14: Instance Termination Confirmation

The EC2 instance is successfully terminated and all associated compute resources are released.





Conclusion

A static website consisting of multiple HTML files was successfully hosted on an Ubuntu-based EC2 instance using the Nginx web server and accessed via a public IPv4